Differential privacy: from statistical decisions to application challenges

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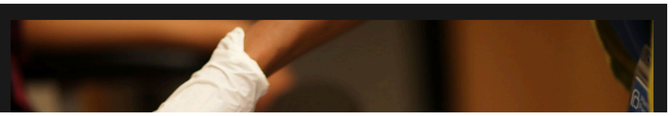
Privacy can be a significant challenge…

… and these are headlines from 3 years go

BAY AREA

Patient's 'embarrassing' private health information posted to Facebook after Contra Costa County medical privacy breach

by: Tori Gaines
Posted: Mar 15, 2023 / 03:10 PM PDT
Updated: Mar 16, 2023 / 05:42 AM PDT



The New York Times

Data Breach Could Compromise Lawmakers' Personal Information

A cyberattack on the District of Columbia's online health insurance marketplace may have compromised identifying data of many members of Congress and other users.

Give this article



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ChatGPT owner OpenAI fixes bug that exposed users' chat histories

According to reports, the titles of the conversations were visible but the substance of other users' conversations was not.

"Perhaps the most striking thing about the right to privacy is that nobody seems to have any clear idea what it is."

Judith Jarvis Thomson, *The Right to Privacy*, Philosophy & Public Affairs 4(4), 1975.

[Al Jazeera, March 23, 2023][NY Times, March 8, 2023][KRON 4, March 15, 2023]

Conceptualizing privacy

“Semantic overload” can cause a lot of confusion

> “We should conceptualize privacy by focusing on the specific types of disruption and the specific activities disrupted rather than looking for the common denominator that links all of them.”
>
> Daniel Solove, *Conceptualizing Privacy*, California Law Review 90(4), 2002.

A lot of things could be called “privacy violations”:

- Targeted ads that are creepy/weird.
- Data brokers aggregating data without consent.
- Hacking, identity theft etc.

This raises important questions:

- Which things are “security” versus “privacy?”
- Which things are “policy” versus “technology?”
- What guarantees can we actually make?

Privacy-preserving machine learning

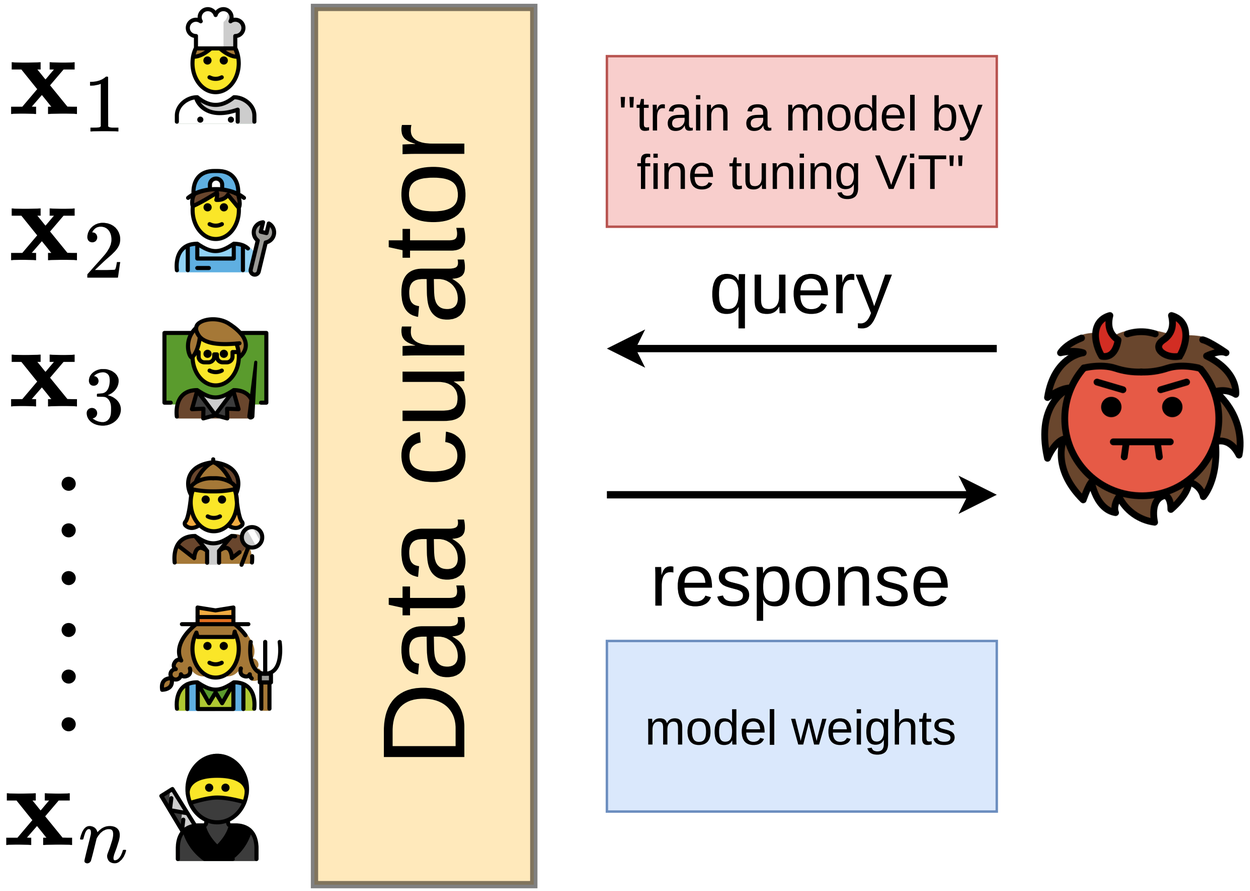
Can we learn from sensitive data without compromising privacy?

A data manager (*the curator*) holds data from multiple individuals.

An outside party (*analyst*) wants to use the data for training a model.

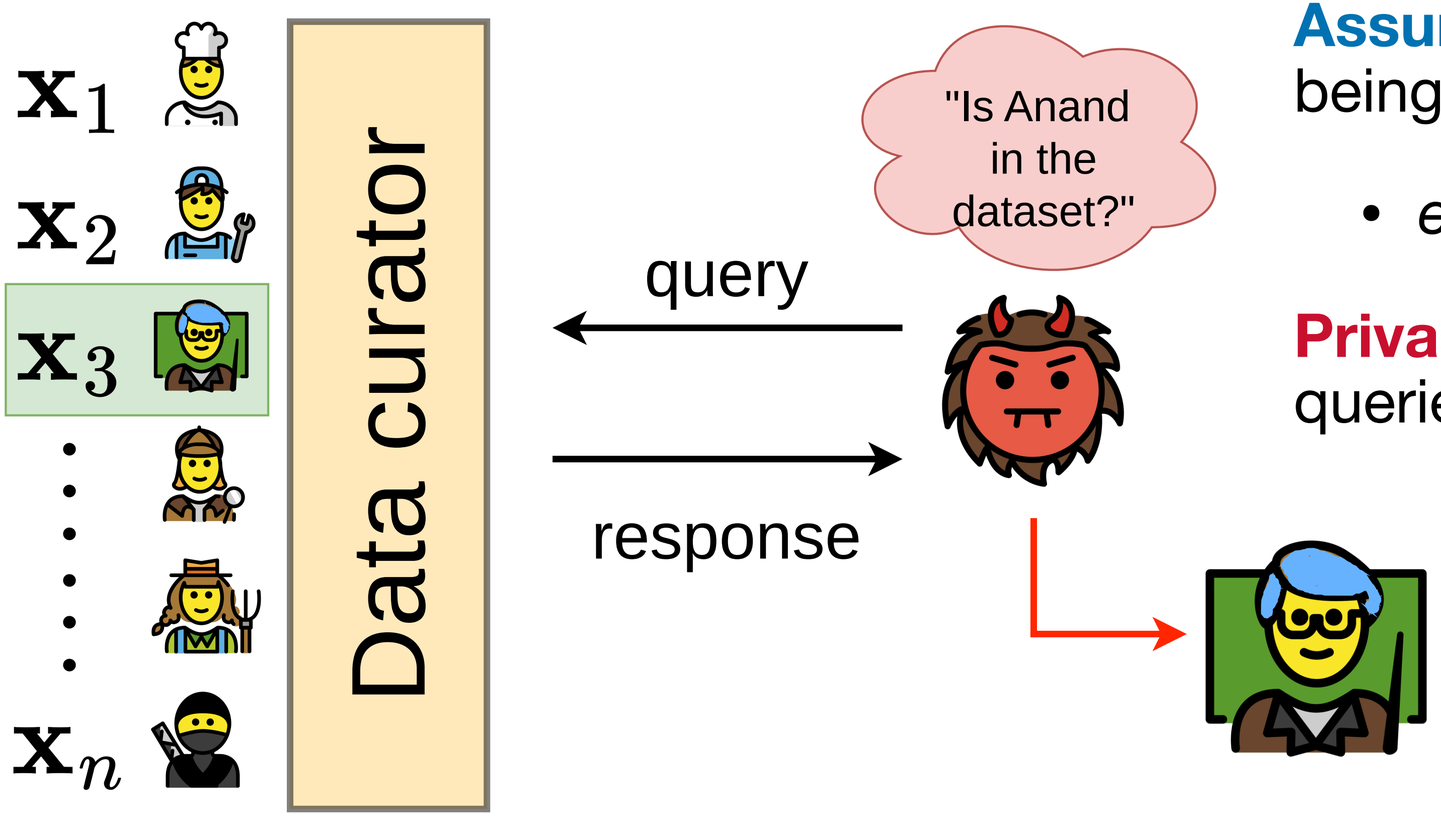
- Individuals' information is *sensitive*.
- Analyst is not trusted: in worst case they are *adversarial*.

What can the curator promise?



Differential privacy (DP) and its guarantee

The goal: protect whether an individual is *in* the dataset



Assumption: individuals care about being *identified as being in the data.*

- *e.g.* participant in a medical study.

Privacy risk comes from answering queries!

- Focus on how the response is *computed*.
- Leakage accrues with more queries.

Differential privacy in practice

It has become the *de facto* "gold standard" for privacy

- 2006: Dwork, McSherry, Nissim, and Talwar propose the basic definition.
- From then onward, a lot of work on *theory* and *algorithms,* especially for machine learning.
- 2014: Erlingsson et al. (Google) proposes RAPPOR for Chrome.
- 2016: Abadi et al. (Google) propose deep learning using DP stochastic gradient descent.
- 2017: Ding et al. (Microsoft) propose DP for telemetry data.
- 2016: Apple announces that it will use DP for device analytics.
- 2020: Abowd et al. (US Census Bureau) uses DP to produce counts for the decennial census.
- 2020: Jin and McGorman (Meta) use DP for COVID tracking.
- 2023: Zhang et al. DP for Google Trends.

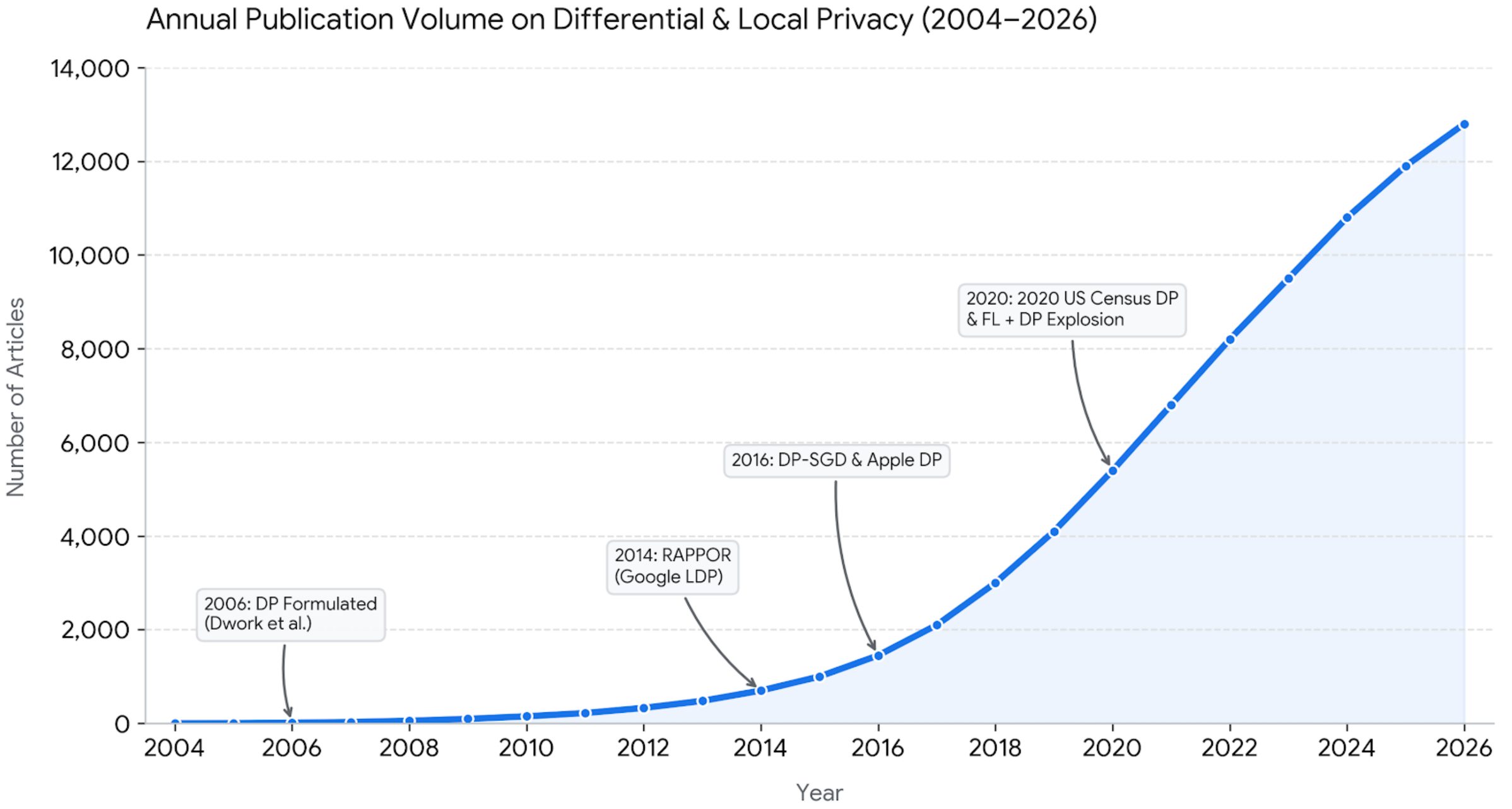
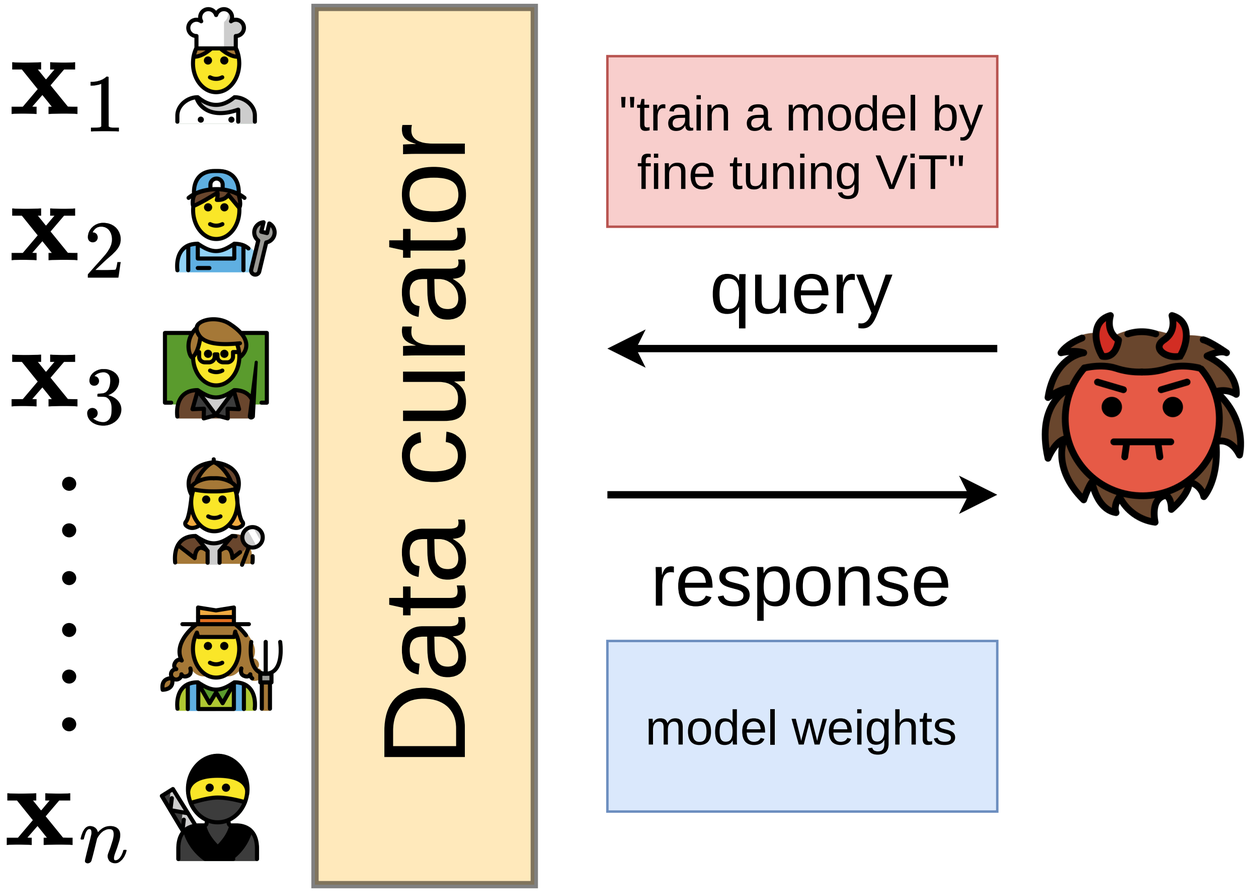


Image generated by Gemini from Google Scholar data

What I'll talk about today

A tour through DP in federated MLSP and a view forward

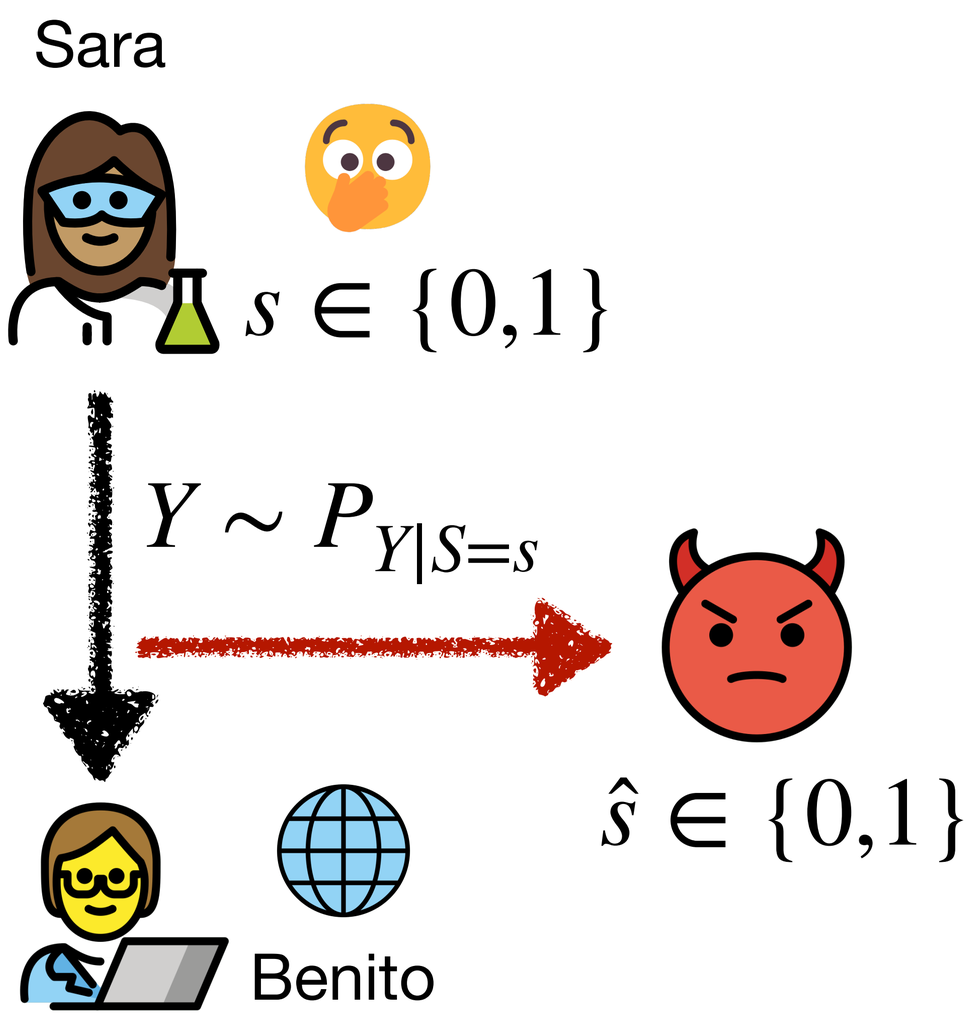


1. Fundamental connections to hypothesis testing.
2. The "textbook" approach to DP.
3. Estimation, learning, and federation.
4. Some thoughts on future directions.

differential privacy
=
hypothesis testing

Privacy at its simplest: protect 1 secret bit

A binary hypothesis test



Sara wants to share information to a collaborator Benito.

- She shares a random variable Y.
- The distribution of Y depends on a **secret** $s \in \{0,1\}$.
- How should she design Y to not reveal s?

The **privacy question** is a **hypothesis testing question**:

$$\mathscr{H}_0 : Y \sim P_{Y|S=0}$$

$$\mathscr{H}_1 : Y \sim P_{Y|S=1}$$

Neyman-Pearson tells us the optimal test

Adversarial inference is a generalized LRT

The optimal test for the adversary is a **likelihood ratio test**:

$$\hat{s}(y) = \begin{cases} 1 & \log \dfrac{P_{Y|S=1}(y)}{P_{Y|S=0}(y)} \geq \tau \\ 0 & \log \dfrac{P_{Y|S=1}(y)}{P_{Y|S=0}(y)} < \tau \end{cases}$$

Example

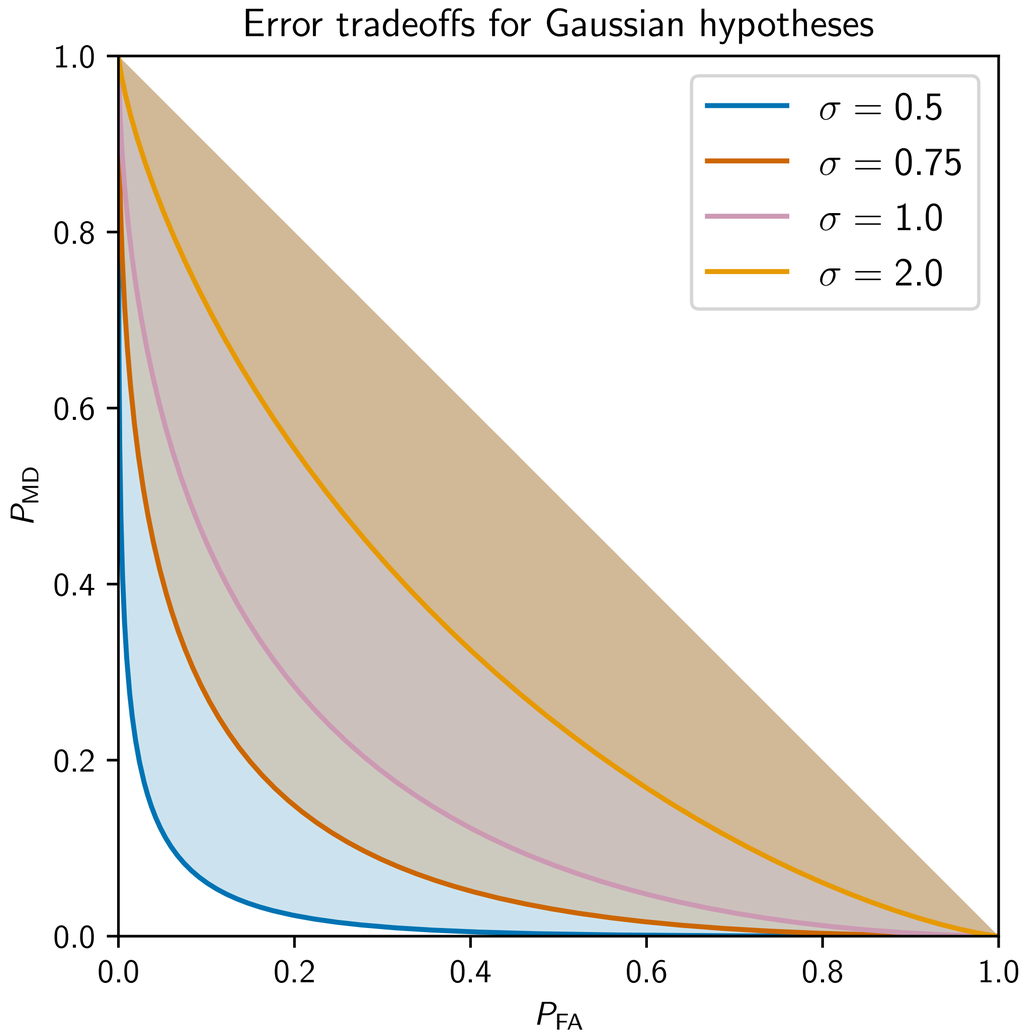
$$\mathcal{H}_0 \colon Y = 0 + Z \sim \mathcal{N}(0, \sigma^2)$$

$$\mathcal{H}_1 \colon Y = 1 + Z \sim \mathcal{N}(1, \sigma^2)$$



Example: additive Gaussian noise

Everyone's favorite example: Gaussians!



If the revealed information Z is **Gaussian**:

$$\mathscr{H}_0 : Z \sim \mathcal{N}(0,\sigma^2)$$

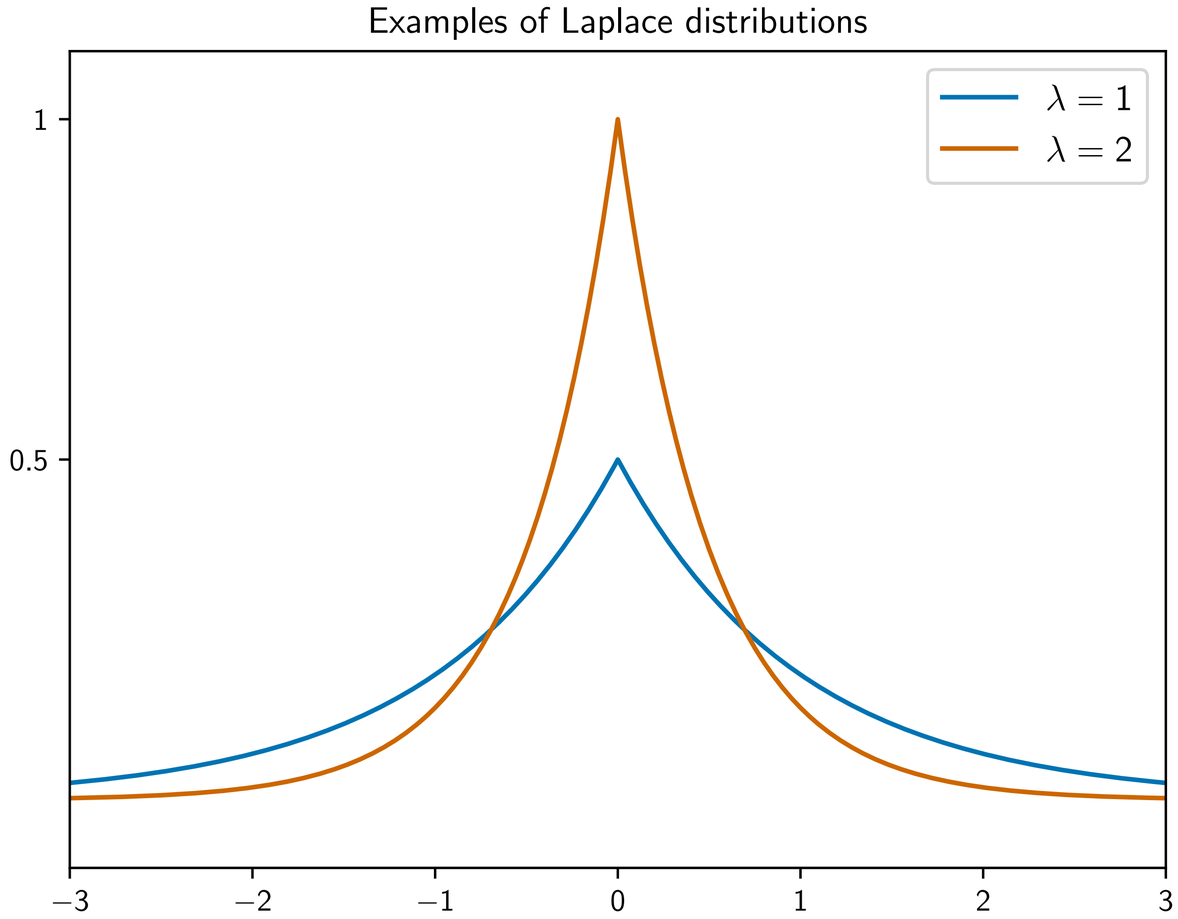
$$\mathscr{H}_1 : Z \sim \mathcal{N}(1,\sigma^2)$$

We can write the error probabilities in terms of Q functions:

$$P_{\text{FA}} = Q\left(\frac{t}{\sigma}\right), P_{\text{MD}} = Q\left(\frac{1-t}{\sigma}\right).$$

Example: additive Laplace noise

We can do more than just Gaussians!



If the revealed information Z is **Laplace**:

$$\mathscr{H}_0 : X \sim \text{Laplace}(\lambda)$$

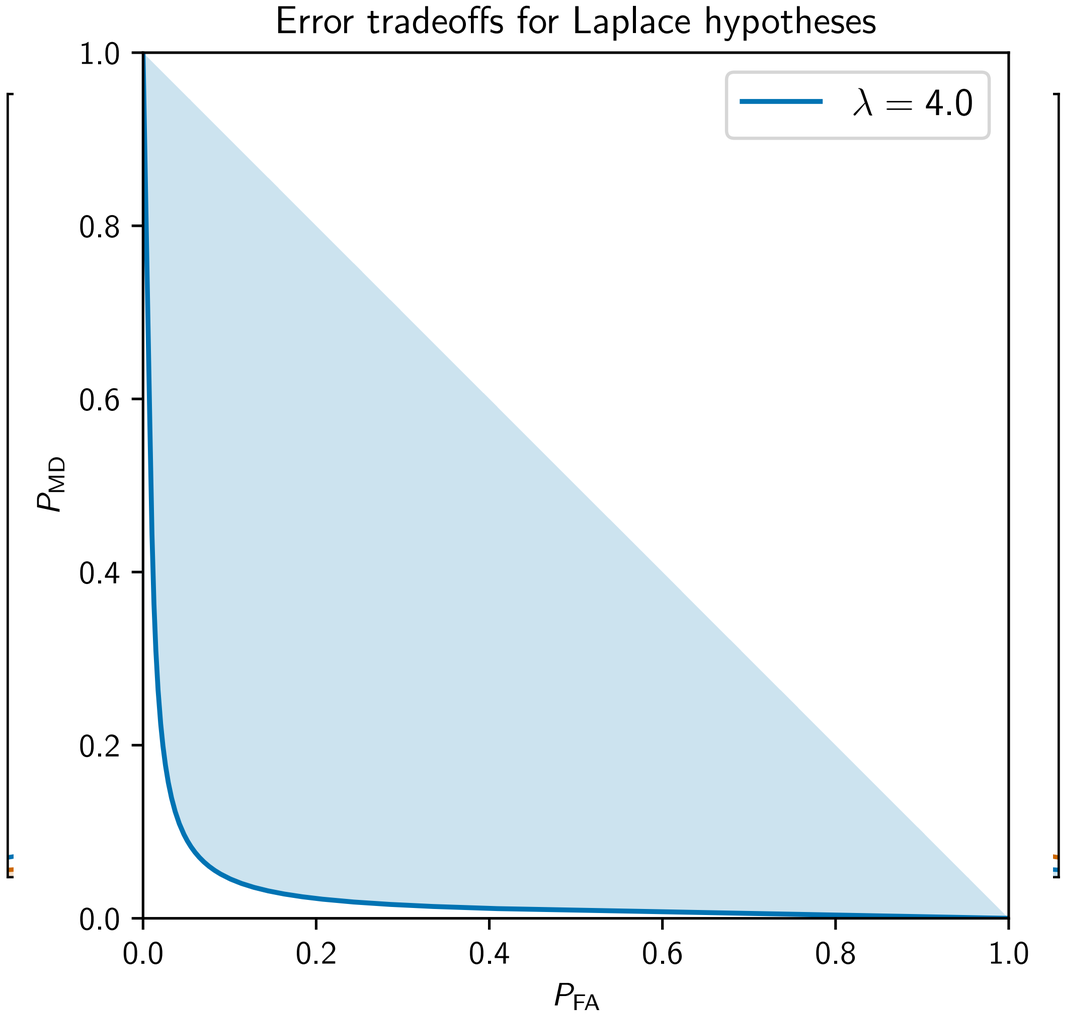
$$\mathscr{H}_1 : X \sim 1 + \text{Laplace}(\lambda)$$

Where Laplace(λ) has density

$$p(z) = \frac{\lambda}{2} \exp(-\lambda\,|\,z\,|\,).$$

Error tradeoffs for Laplace noise

Lighter tails give a different shape



The error probabilities for the test are:

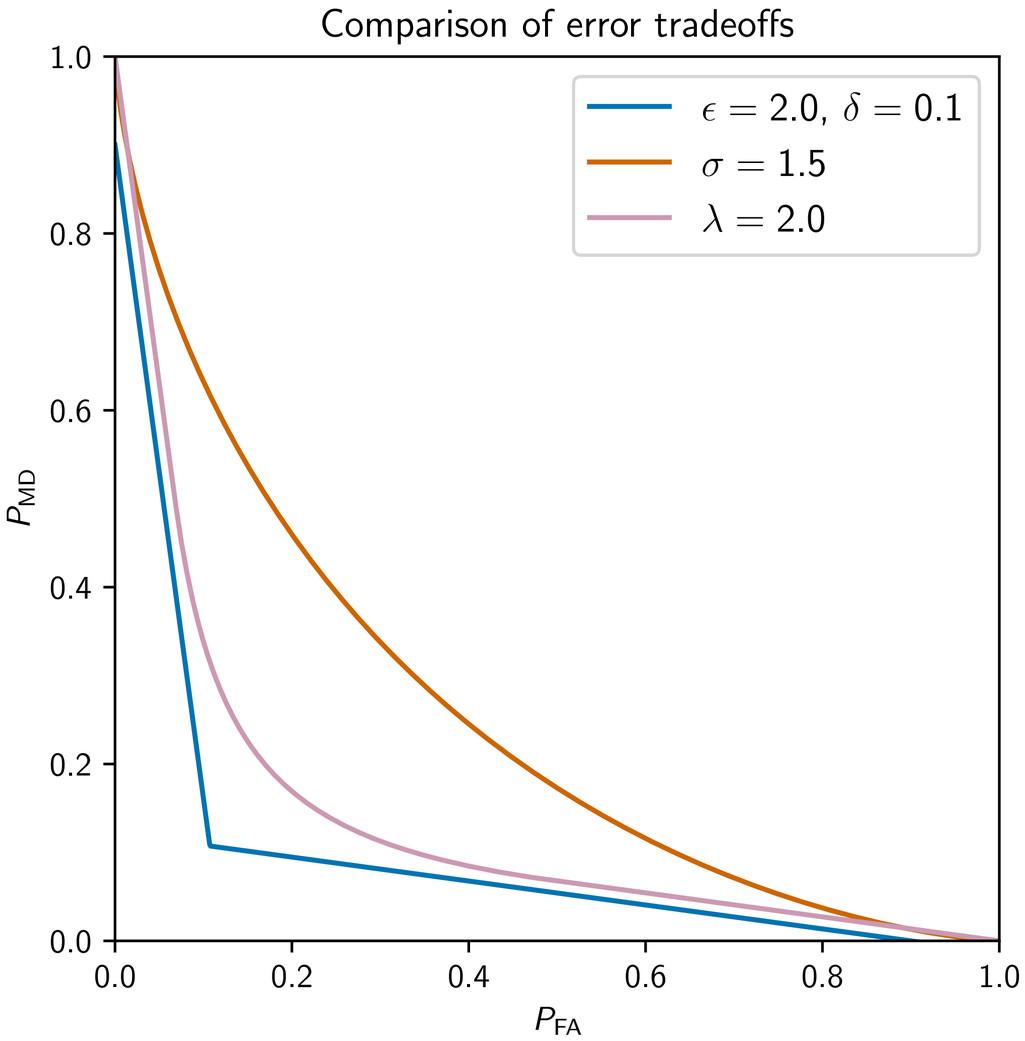
$$P_{\text{FA}} = \int_{t}^{\infty} \frac{\lambda}{2} \exp(-|t|\lambda) dt$$

$$P_{\text{MD}} = \int_{-\infty}^{t} \frac{\lambda}{2} \exp(-|A-t|\lambda) dt$$

The tradeoff is similar to the Gaussian but the slope at the corners is different.

Tradeoffs between P_{FA} and P_{MD}

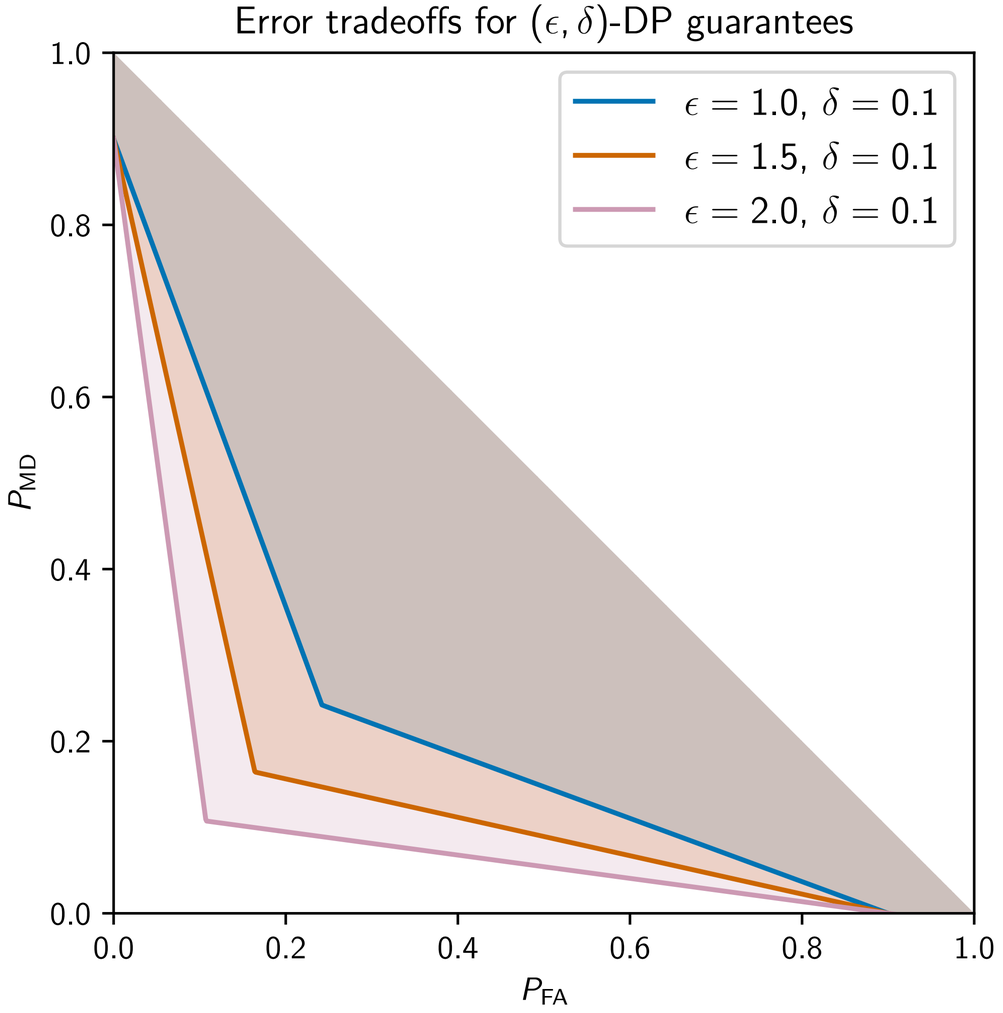
We get more privacy when the hypothesis test is “hard”



- A **privacy guarantee** is made by the tradeoff between probabilities of
  - **false alarm** (Type I error) and
  - **missed detection** (Type II error)
- If the likelihood ratio is small, the test will have a higher error.
- We can use this “flipped” ROC curve to visualize the tradeoff.

Tradeoffs between P_{FA} and P_{MD}

Commonly used noise distributions do not meet the lower bound



A **privacy guarantee** is made by the tradeoff between probabilities of **false alarm** (Type I error/false positive) and **missed detection** (Type 2 error/false negative).

The DP definition prescribes a **piecewise linear lower bound** on the error tradeoff:

$$P_{\text{FA}} + e^{\epsilon} P_{\text{FA}} \geq 1 - \delta$$

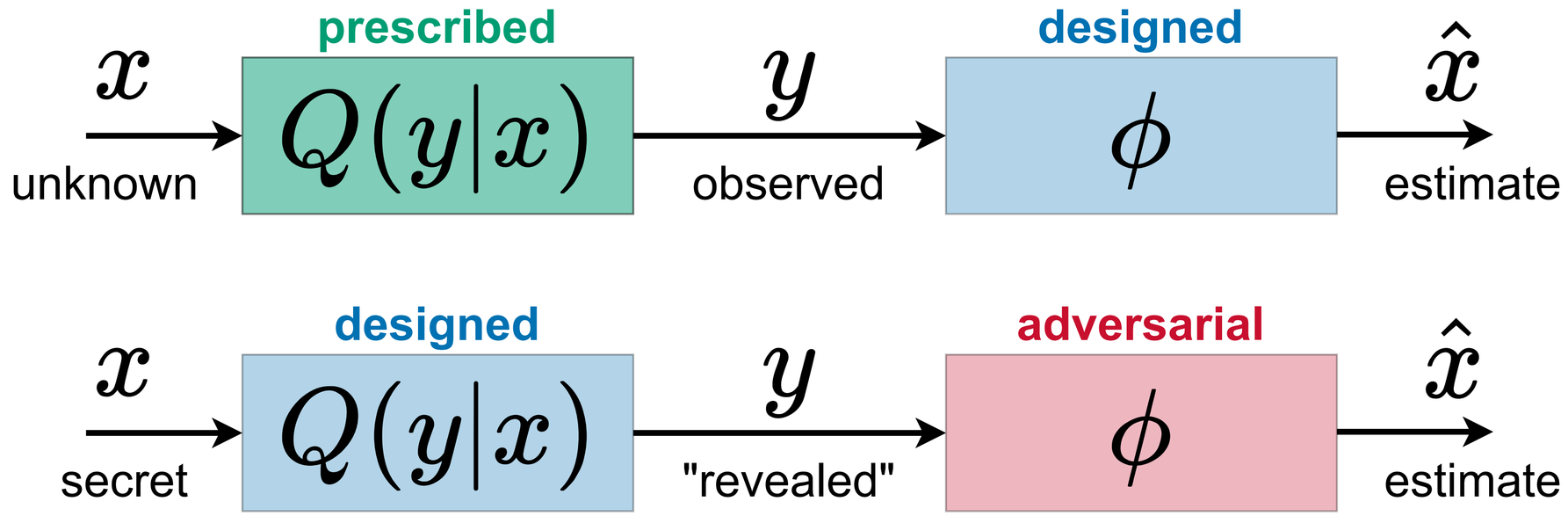
$$e^{\epsilon} P_{\text{FA}} + P_{\text{MD}} \geq 1 - \delta$$

Wasserman, Zhou (2010) Kairouz, Oh, Vishwanath (2015)

Privacy versus testing

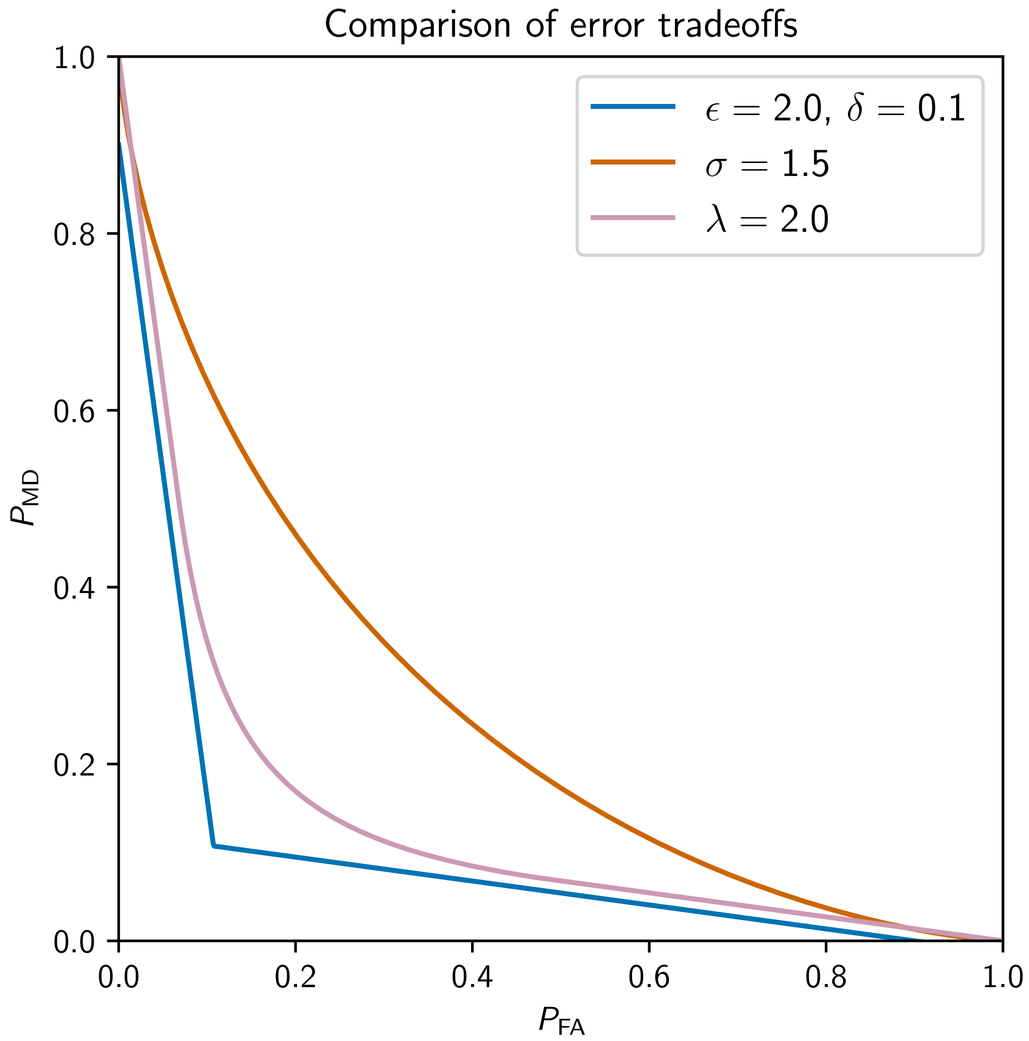
We get to design the test!

The key difference between **hypothesis testing** (as we usually encounter it) and **(differential) privacy** is that we design the **likelihoods** but not the **test**!



Comparing hypothesis tests

Asking if one tradeoff curve is above another



We can sometimes compare two hypothesis testing problems by **comparing the optimal tradeoff curves**.

$$\mathscr{H}_0 \colon Y \sim \mu \qquad \mathscr{H}_1 \colon Y \sim \nu$$

$$\mathscr{H}'_0 \colon Y \sim \mu' \qquad \mathscr{H}'_1 \colon Y \sim \nu'$$

If the curve for the test (μ', ν') is above the curve for (μ, ν) then the test (μ', ν') is "harder" = more private:

$$T(\mu, \nu) \leq T(\mu', \nu')$$

Dong, Roth, Su (2022), Blackwell (1950/51/53), Raginsky (2011)

differential privacy

“the normal way”

The “standard” approach to explaining DP

Neighboring databases of individual records

In the textbook approach to describing DP we have several ingredients:

1. **Data space:** $\mathscr{X}$, often modeled as records from n individuals.

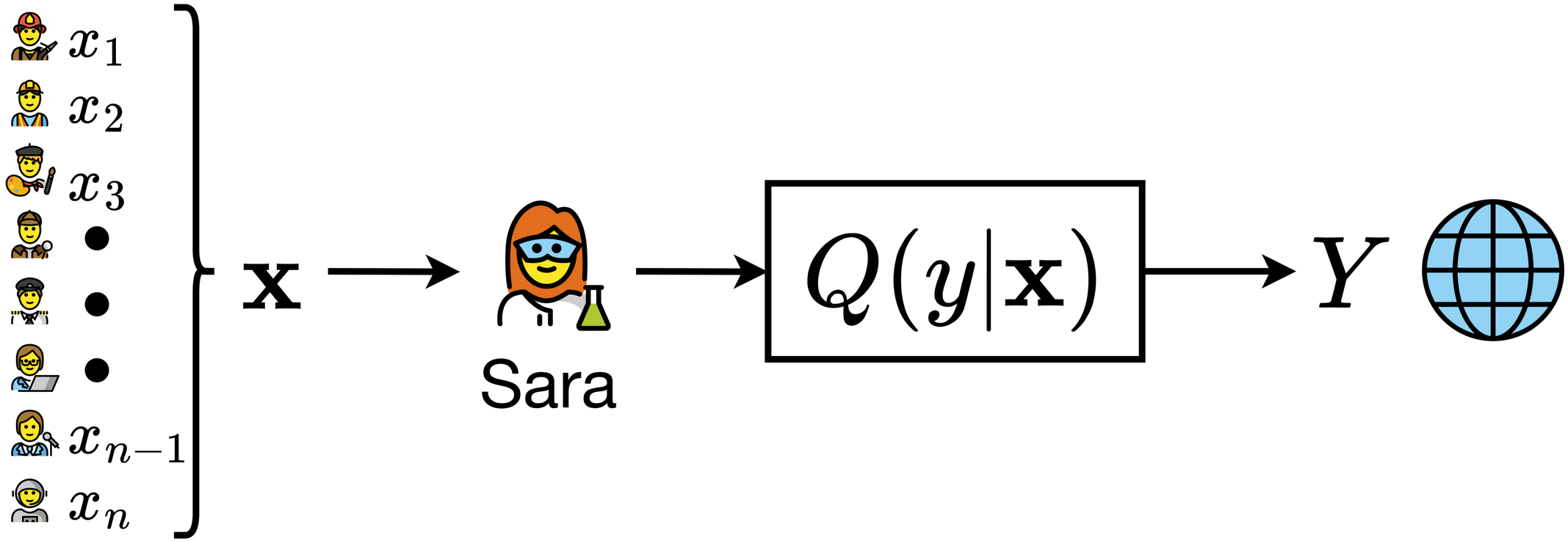
2. **Neighborhood relationship $\sim$:** for $\mathbf{x}, \mathbf{x}' \in \mathscr{X}$ we write $\mathbf{x} \sim \mathbf{x}'$ if they are “neighbors”.
   - Example: each person has 1 bit so $\mathscr{X} = \{0,1\}^n$ and $\mathbf{x} \sim \mathbf{x}'$ if they differ in one position.

3. **Output space:** $\mathscr{Y}$, depends on the functionality/what we want to release.
   - Example: If we want the average of data $\mathscr{X} = [0,1]^n$, we have $\mathscr{Y} = [0,1]$.
   - Example: If we want to train a classifier using data $\mathscr{X} = \{\mathbb{R}^d \times \{0,1\}\}^n$, $\mathscr{Y} = \mathbb{R}^d$.

4. **Algorithm:** a randomized map/conditional distribution/channel $Q\colon \mathscr{X} \to \mathscr{Y}$.

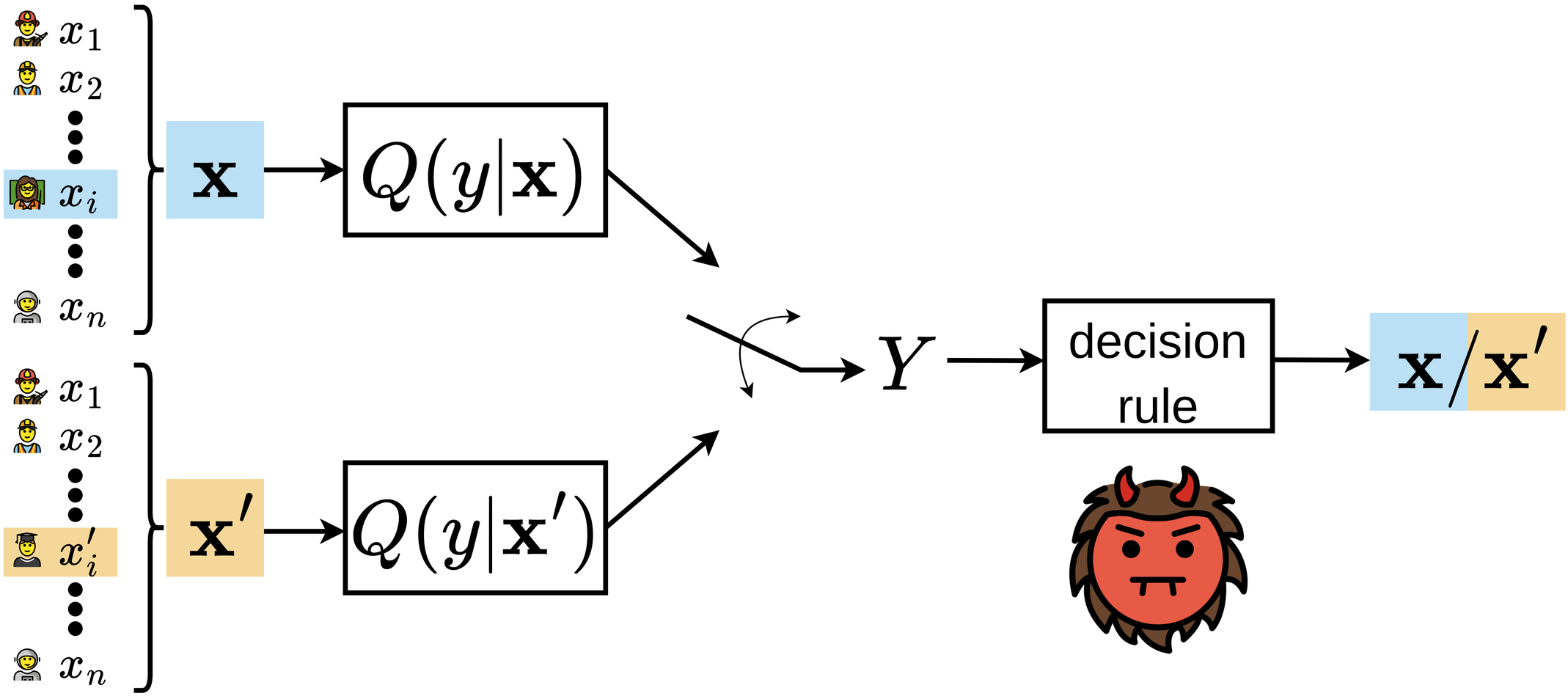
In a snapshot

Replacing a single bit with a database



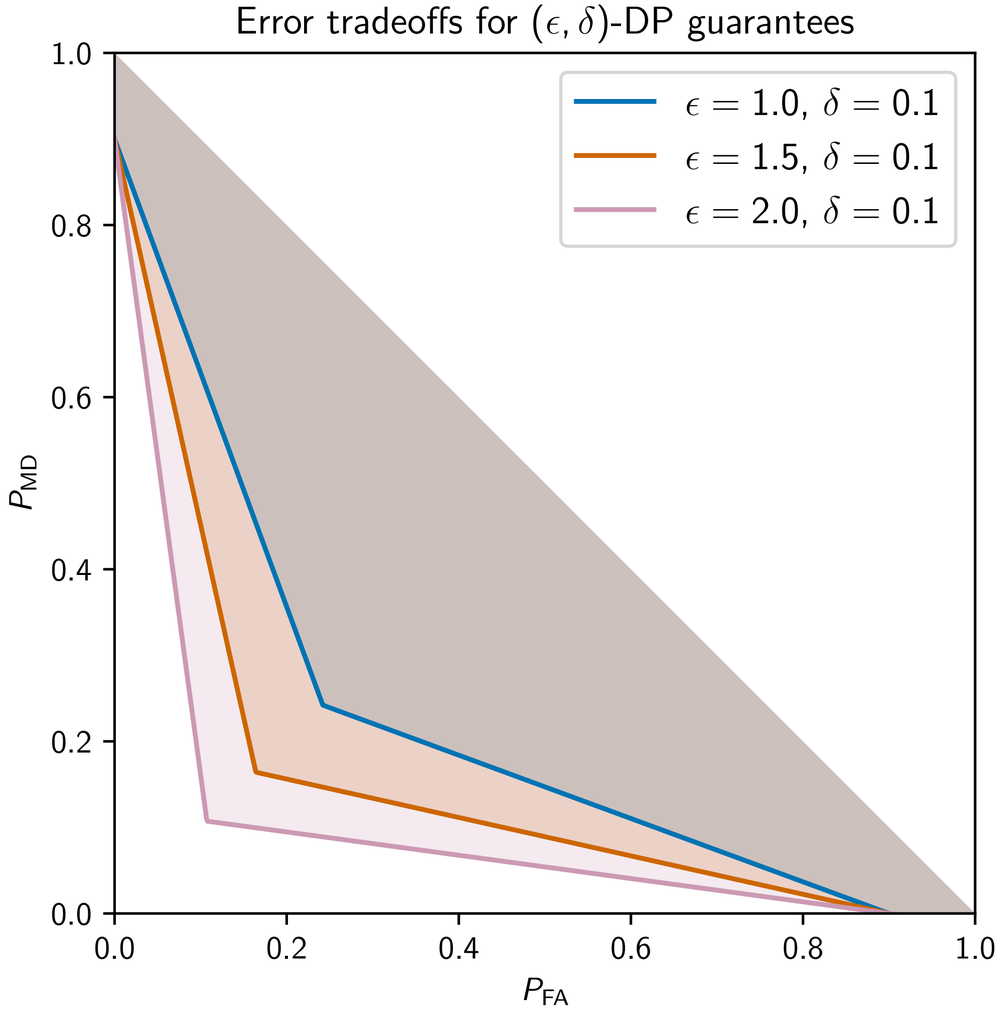
Neighboring datasets in a picture

The adversary's hypothesis test



DP makes many hypothesis tests hard

Protecting many single bits simultaneously



Compared to our single private bit s, in DP we want **many hypothesis tests to hard for the adversary**. For every $\mathbf{x} \sim \mathbf{x}'$ the test

$$\mathscr{H}_0 \colon \mathbf{y} \sim Q(\,\cdot\,|\,\mathbf{x})$$

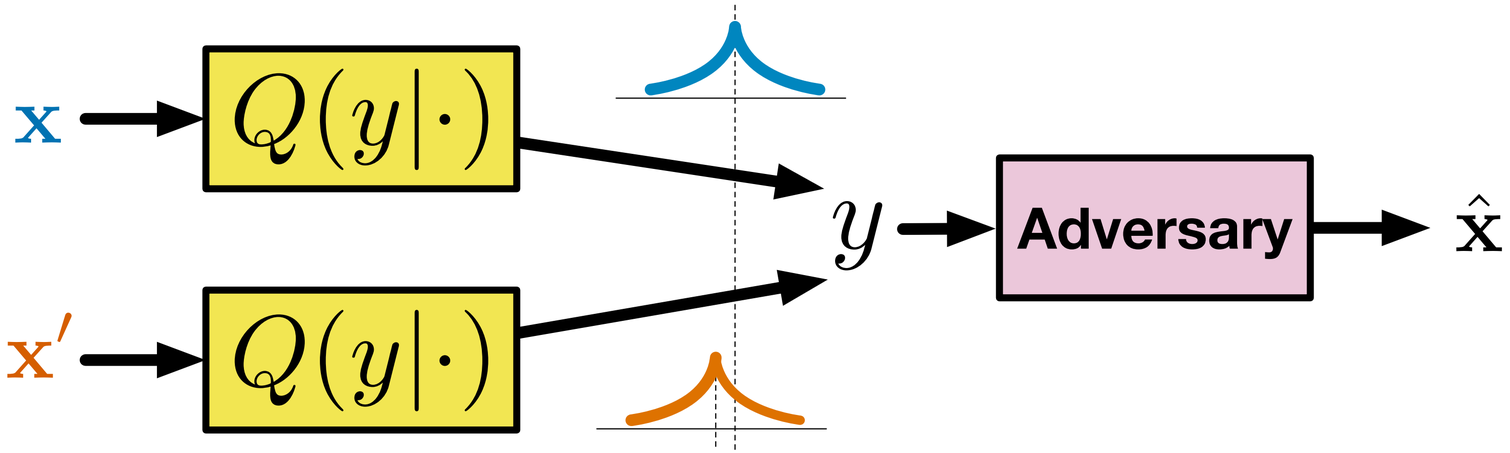
$$\mathscr{H}_1 \colon \mathbf{y} \sim Q(\,\cdot\,|\,\mathbf{x}')$$

should have a large probability of error.

When can we do this? When **neighboring data sets make similar output distributions**.

The hypothesis testing in DP

DP is a property of the channel



A channel/"mechanism"/algorithm Q is (ϵ, δ)**-differentially private** if

$$Q(\mathcal{T} \,|\, \mathbf{x}) \leq e^{\epsilon} Q(\mathcal{T} \,|\, \mathbf{x}') + \delta$$

For all measurable subsets $\mathcal{T} \subseteq \mathcal{Y}$ and all $\mathbf{x} \sim \mathbf{x}'$.

[Dwork-Kenthapadi-McSherry-Mironov-Naor 2006]

[Wasserman-Zhou 2010]

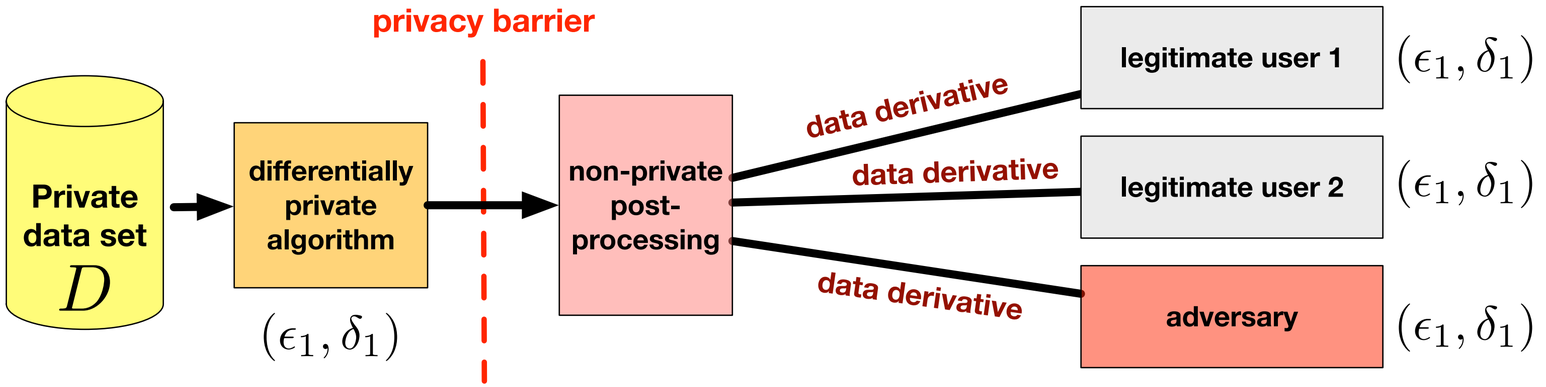
Some notes on the definition

DP's underlying assumptions are slightly different

- **Differential privacy is a pretty stringent requirement:** The probability of any event is similar under $\mathbf{x}$ and any other neighboring $\mathbf{x}'$.
- **Guarantee is on conditional probabilities given the data:** same risk holds regardless of side information (e.g. linkage attacks).
- **There is no statistical assumption on the data:** $\mathbf{x}$ is not drawn from some (prior) distribution.
- **The data itself is considered identifying:** no notion of some parts being personally identifiable information (PII) and others not.

Post-processing invariance and composition

Nice properties of differential privacy



- **Side-information resilience:** measures the additional risk regardless of what is known already.
- **Post-processing invariance:** once we publish something the risk cannot increase from additional computations.
- **Composition:** quantifies how privacy loss “adds up” over multiple releases.

Sensitivity of scalar functions

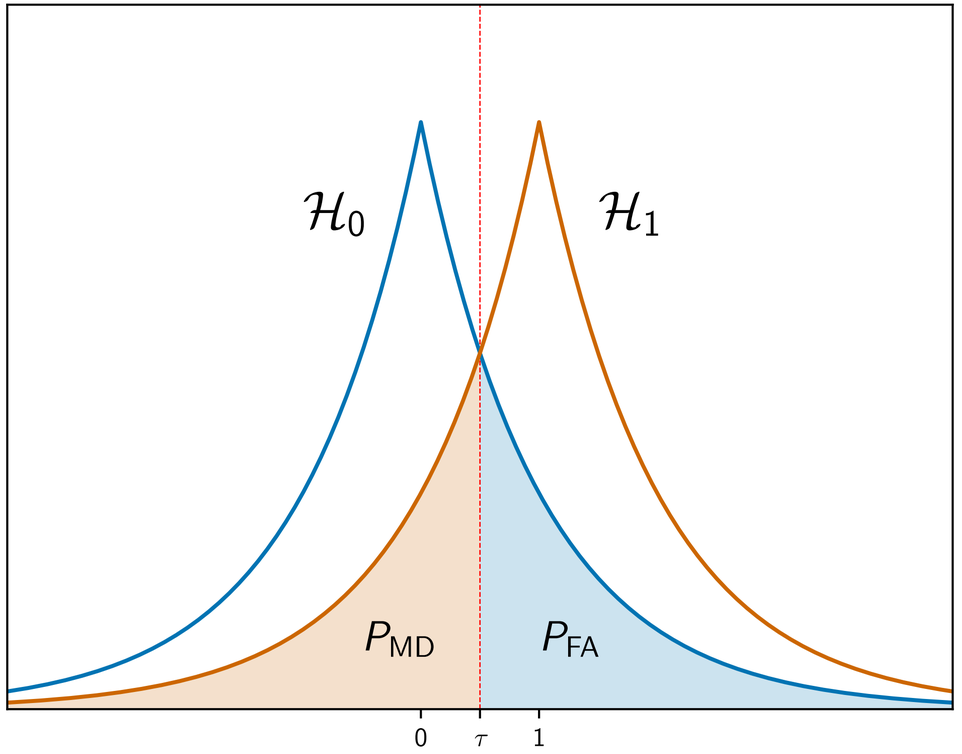
Understanding the distance between hypotheses

In DP, we usually want to approximate some function of the data.

Suppose we want $f : \mathcal{X} \to \mathbb{R}$. We want the test to be hard for any pair $(\mathbf{x}, \mathbf{x}')$ which are "neighbors" $(\mathbf{x} \sim \mathbf{x}')$.

If $f(\cdot)$ is small for all neighbors, this should be easier.

Example: $f(x) = \frac{1}{n}\sum_{i=1}^{n} x_i$ can change by at most $\frac{1}{n}$ for $x_i \in [0,1]$.



Sensitivity of scalar functions

Understanding the distance between hypotheses

The **global sensitivity** of a scalar function $f(\cdot)$ is

$$\Delta(f) = \max_{\mathbf{x} \sim \mathbf{x}'} \left| f(\mathbf{x}) - f(\mathbf{x}') \right|.$$

If we use additive noise (like in the Laplace and Gaussian case) we have

$$\mathscr{H}_0 \colon Z \sim p(z - f(\mathbf{x})) \qquad \text{vs.} \qquad \mathscr{H}_1 \colon Z \sim p(z - f(\mathbf{x}'))$$

We can make a guarantee for all “neighbors” if following test is hard:

$$\mathscr{H}_0 \colon Z \sim p(z) \qquad \text{vs.} \qquad \mathscr{H}_1 \colon Z \sim p(z - \Delta(f)).$$

Example: the sample mean

Computing the MSE as a function of privacy risk

Suppose we have data in $\mathcal{X} = [A, B]^n$ and want to estimate the mean:

$$\hat{\mu}(\mathbf{x}^n) = \frac{1}{n}\sum_{j=1}^{n} \mathbf{x}_j + Z$$

- Sensitivity of $\hat{\mu}(\mathbf{x}^n)$ is $(B - A)/n$.
- $Z \sim$ Laplace$(n\epsilon/(B - A))$ will guarantee $(\epsilon,0)$-DP.
- MSE of $\hat{\mu}(\mathbf{x}^n)$ is $2/\lambda^2 = 2\dfrac{(B - A)^2}{n^2\epsilon^2}$.

The privacy-utility tradeoff

How much do we lose when we guarantee privacy?

Adding Laplace(λ) noise guarantees privacy, but at what cost? The MSE is:

$$2/\lambda^2 = 2\frac{(B-A)^2}{n^2\epsilon^2}$$

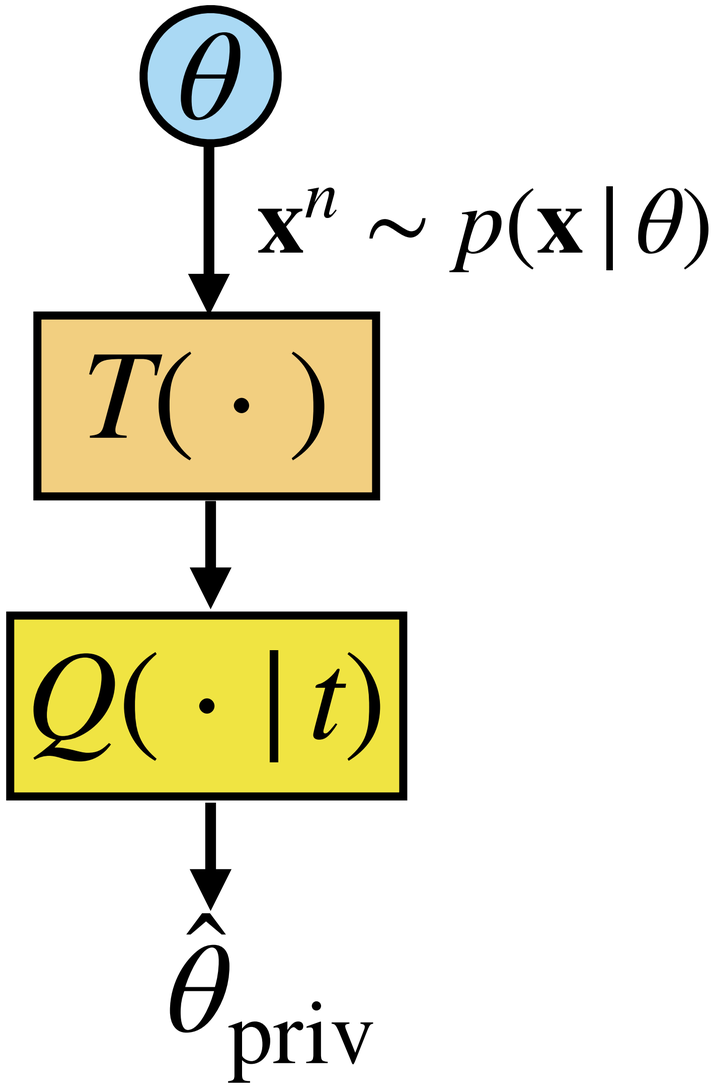
So we can see that **less privacy risk (smaller ϵ) induces more MSE**.

We can try to optimize the privacy mechanism if we know the utility function (like squared error).

This is what people call the **privacy-utility tradeoff**.

Point estimation with differential privacy

Adding noise to sufficient statistics



A typical DP approach to statistical estimation (Smith 2009):

- Model data as drawn i.i.d. $\sim p(\mathbf{x} \mid \theta)$.
- Compute a sufficient statistic $T(\mathbf{x}^n)$ for θ.
- Add noise to $T(\mathbf{x}^n)$ to guarantee DP.
- Compute a "plug-in" estimate from noisy $T(\mathbf{x}^n)$.

All we need to know is the sensitivity of $T(\cdot)$.

What kind of noise should we use?

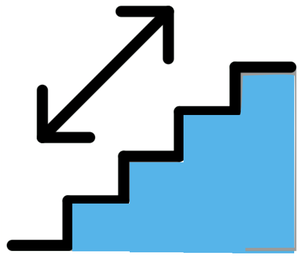
So many different choices: a non-comprehensive list

Variations on geometric noise

Ghosh, Roughgarden, Sundarajan (2009/2012)

Gupte, Sundararajan (2010)

Balcer, Vadhan (2017)



Variations on staircases

Geng, Viswanath (2015, 2016

Kairouz, Oh, Viswanath (2014)

Geng, Kairouz, Oh, Viswanath (2015)

Kalantari, Sankar, Sarwate (2018)



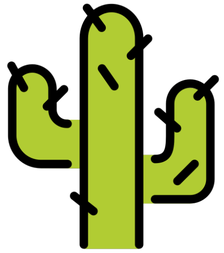
Variations on Gaussians

Cannone, Kamath, Steinke (2020)

Sadeghi, Korki (2022)

Muthukrishnan, Kalyani (2023)

Rinberg, Shumailov, Cummings, Papernot (2024)



“Other"

Geng, Ding, Guo, Kumar (2019/2020)

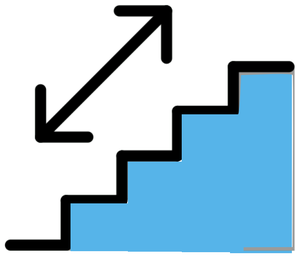
Dong, Su, Zhang (2021)

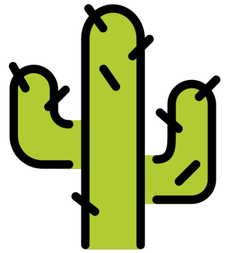
Alghamdi, Asoodeh, Calmon, Kosut, Sankar, Wei (2022)

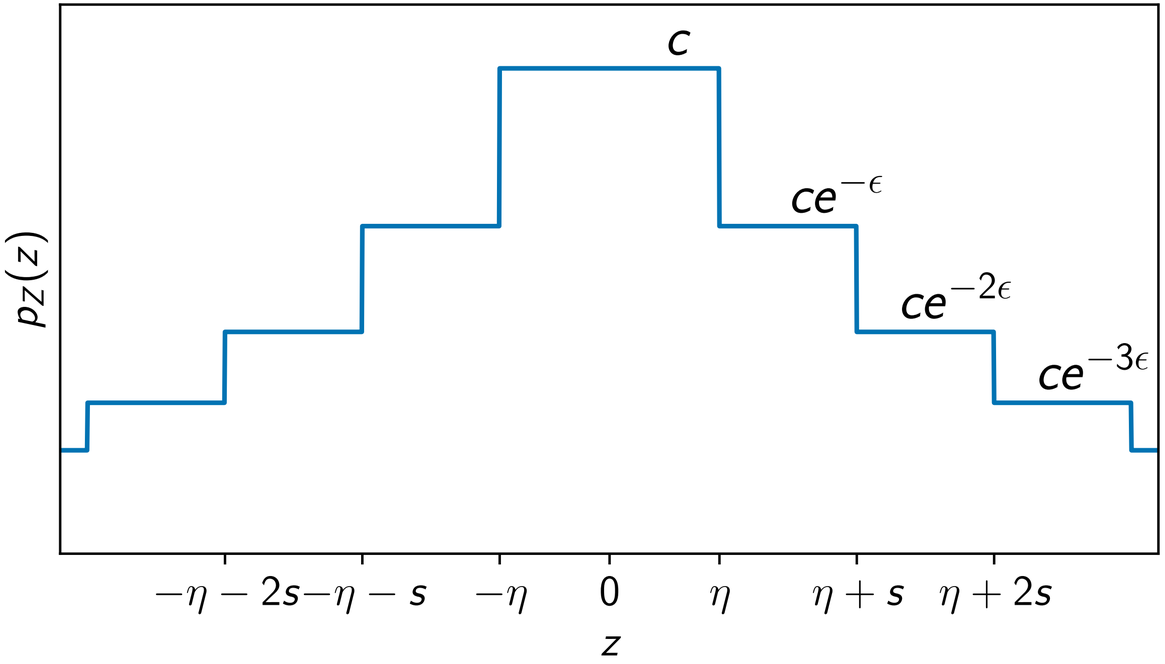
Alghamdi, Asoodeh, Calmon, Gomez, Kosut, Sankar (2023)

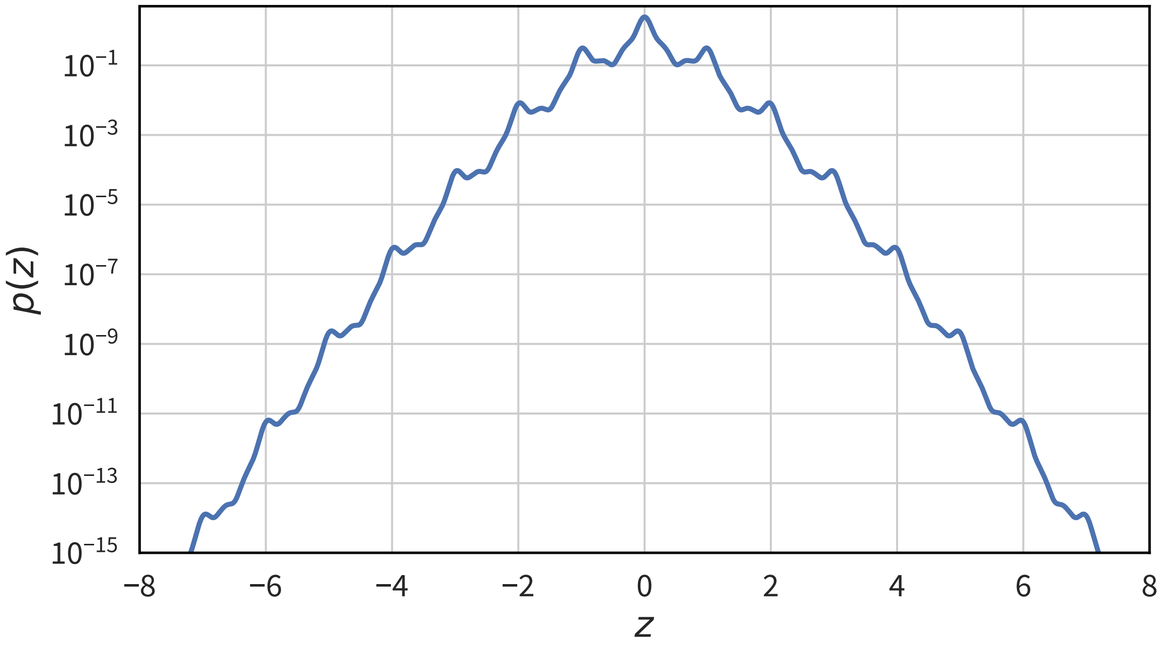
“Optimal” noise distributions

Beyond Gaussian and Laplace









Beyond additive noise

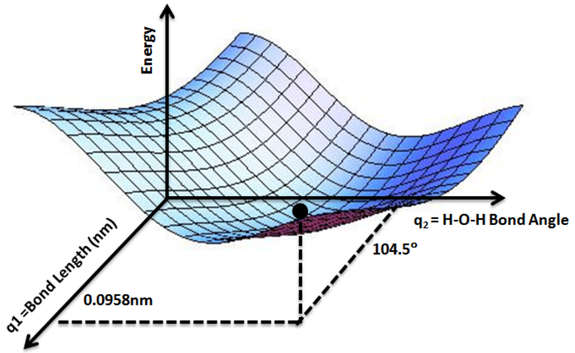
Differential privacy requires *randomization*, not just noise

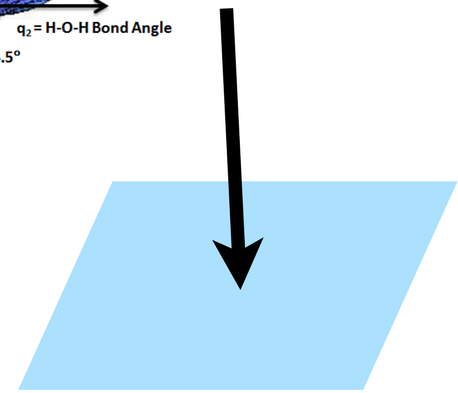
Other types of randomization can also guarantee DP:

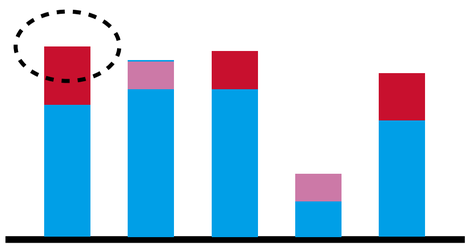
- **Sampling from a Gibbs distribution** (the "exponential mechanism"):

$$Q(\mathbf{y} \mid \mathbf{X}) \propto \exp\left(\epsilon \cdot U(\mathbf{X}, \mathbf{y})\right)$$

- **Random projection**
- **Selection from noisy alternatives**







Surface image from AimNature (Wikipedia)

differential privacy
+
federated MLSP

Maximum likelihood and ERM

Optimization and privacy

Most ML/AI training (and classical stats) uses optimization to minimize some loss:

$$\mathbf{w}^* = \operatorname*{argmin}_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} \ell(\mathbf{w}, \mathbf{x}_i).$$

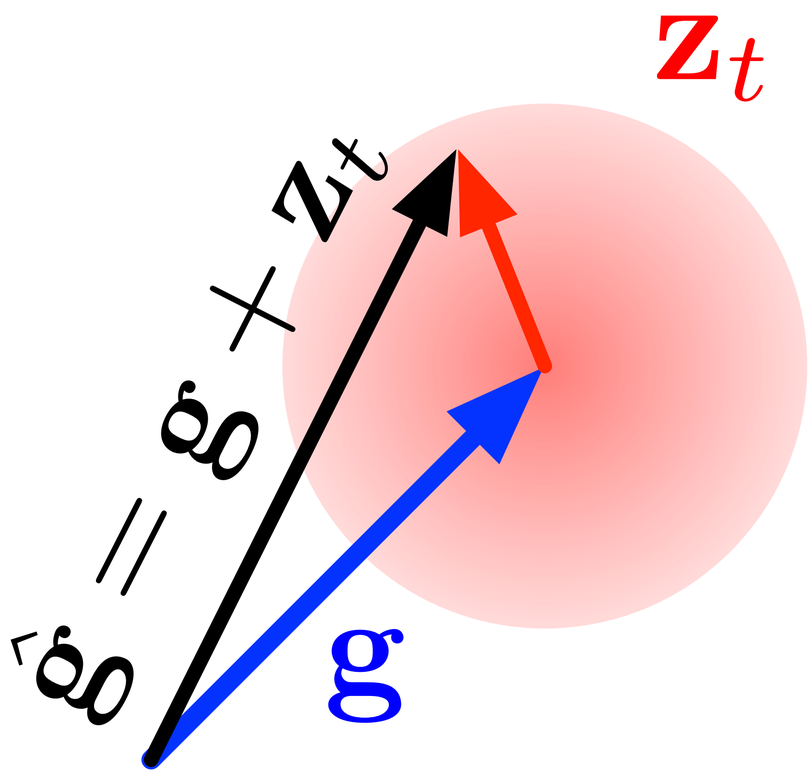
We can use DP to approximate this in a number of ways:

- **"Output perturbation"**: compute the minimizer and add noise.
- **"Objective perturbation"**: Add a random term to the objective function and minimize it.
- **"Functional mechanism"**: Add noise to an approximation of the loss function $\ell(\,\cdot\,)$.

[Chaudhuri, Monteleoni, Sarwate 2011] [Zhang, Zhang, Xiao, Yang, Winslett 2012]

Deep Learning and DP

Privacy for neural networks



Deep neural networks (DNNs) use stochastic optimization algorithms in training: we can make stochastic gradient descent (SGD) differentially private by **adding noise to the gradients**.

- For high-dimensional problems, Gaussian noise is very effective.
- SGD has many iterations so we are potentially leaking a lot of information.

Privacy accounting lets us track the overall privacy loss: these are **composition analyses**.

[Song et.al. 2013, Duchi et.al. 2014, Abadi et.al. 2016, Mironov 2017]

Privacy loss random variables

Characterizing the distribution of the LLR

The fundamental quantity of interest is called the **privacy loss random variable** (PLRV). Switching notation a little bit, we can define the variable

$$L_{x,x'} = \log \frac{dP_{Y|X=x}}{dP_{Y|X=x'}}(Y)$$

$$\mathbb{E}[L] = D_{\mathrm{KL}}(P_{Y|x} \| P_{Y|x'})$$

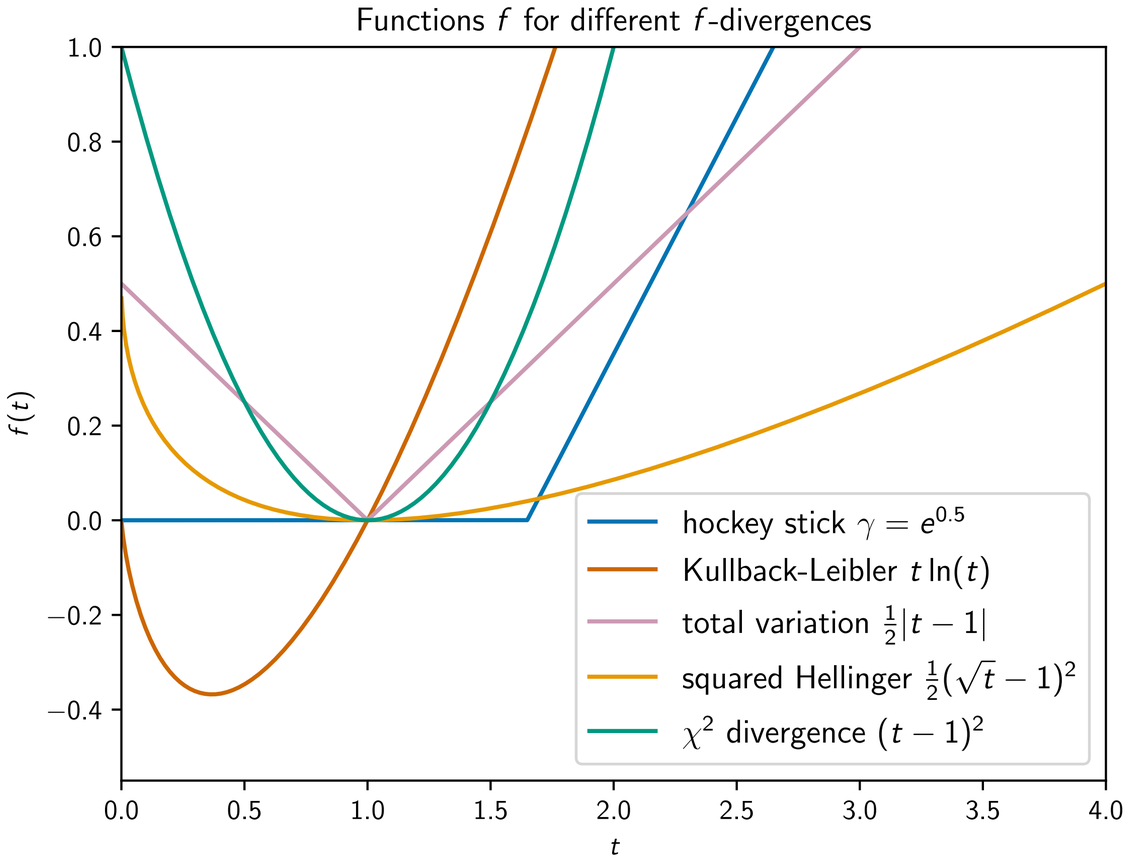
The distribution of the PLRV is sometimes called the **privacy loss distribution** (PLD).

A challenge: this is defined for a single pair of inputs (x, x'). We would like to only deal with the "worst case" pair of inputs.

Sommer, Meisner, Mohammadi (2020), Zhu, Dong, Wang (2022)

Generalized divergences and the 🏒 divergence

How different are these two distributions?



More generally, look at f-divergences: for any convex $f(\cdot)$,

$$D_f(\mu\|\nu) = \mathbb{E}_\nu\left[f\left(\frac{d\mu}{d\nu}\right)\right].$$

If we choose $f(t) = (t-\gamma)^+$ we get the "hockey stick", or E_γ divergence:

$$\mathsf{E}_\gamma(\mu\|\nu) = \int_\Omega \left(\frac{d\mu}{d\nu} - \gamma\right)^+ d\nu = \sup_A \left[\mu(A) - \gamma\nu(A)\right]$$

Rényi (1961), Cziszár (1963), Morimoto (1963), Ali, Silvey (1966), Csiszár (1967), Polyanskiy, Poor, Verdu (2010), Balle, Barthe, Gaboardi, Geumlek (2019)

Composition and divergences

Adding things up

If we have multiple DP computations with PLRVs $L_1, L_2, \ldots, L_k$:

$$L_{\text{total}} = L_1 + L_2 + \cdots L_k.$$

Building from measure concentration (Dwork, Rothblum, Vadhan (2010)) or exact composition (Kairouz, Oh, Viswanath (2015), Murtagh, Vadhan (2016)) there are (at least) three main branches of work on composition:

- Large deviations
- Central limit theorem
- Numerical approaches

With thanks to Flavio Calmon, Oliver Kosut, and Shahab Asoodeh!

A taxonomy of composition analyses

(on non-interactive settings, also non-exhaustive)

Large deviations

Bun and Steinke (2016)

Abadi, Chu, Goodfellow, McMahan, Mironov, Talwar, Zhang (2016)

Mironov (2017)

Wang, Balle, Kasiviswanathan (2018)

CLT and characteristic functions

Dong, Roth, Su (2019)

Sommer, Meiser, Mohammadi (2020)

Cannone, Kamath, Steinke (2020)

Zhu, Dong, Wang (2022)

Numerical methods

Koskela, Jälkö, Prediger, Honkela (2021/2021)

Gopi, Lee, Wutschitz (2021)

Ghazi, Kamath, Kumar, Manurangsi (2022)

Doroshenko, Ghazi, Kamath, Kumar, Manurangsi (2022)

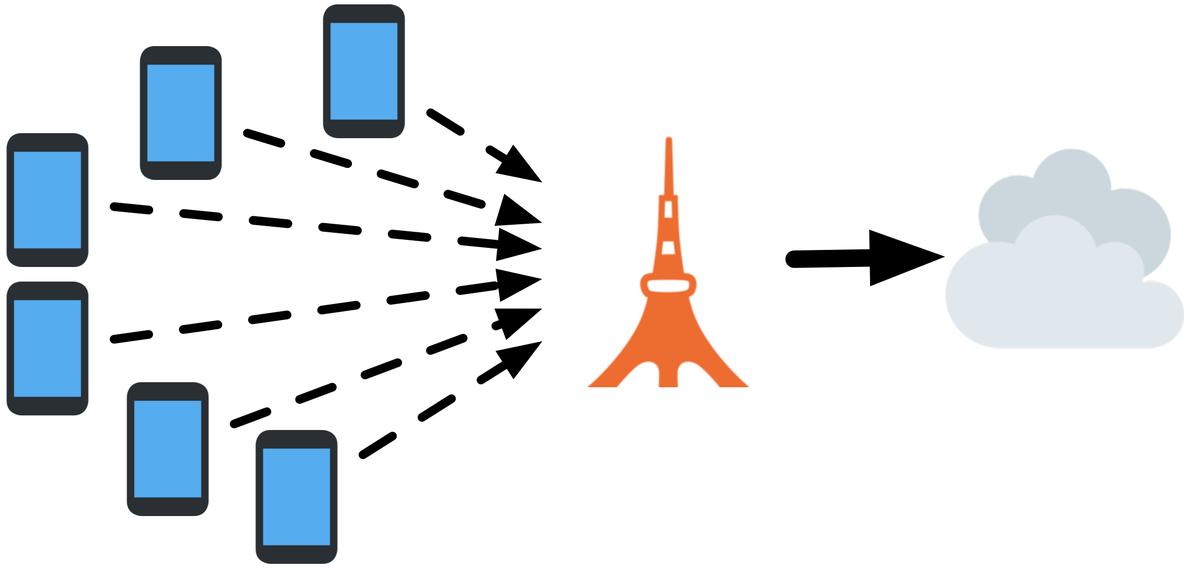
Warning about subsampling!

Lebeda, Regehr, Kamath, Steinke (2024)

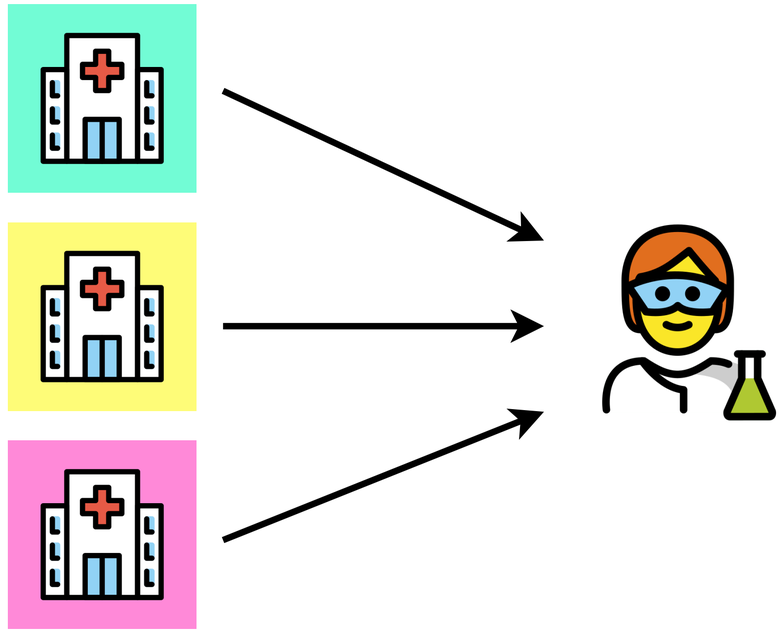
Chua, Ghazi, Kamath, Kumar, Manurangsi, Sinha, Zhang (2024)

Federated learning and privacy

Different models lead to different considerations



Learning from millions of phones



Learning from a few “silos”

What about federated learning *systems*

We often focus on one-shot training: real systems are harder



A federated learning platform for neuroimaging.

- Data holders can maintain local data governance.
- Support joint analyses across public and private data.
- Incentivize allowing access to data through queries alone.

https://neuroflame.org/

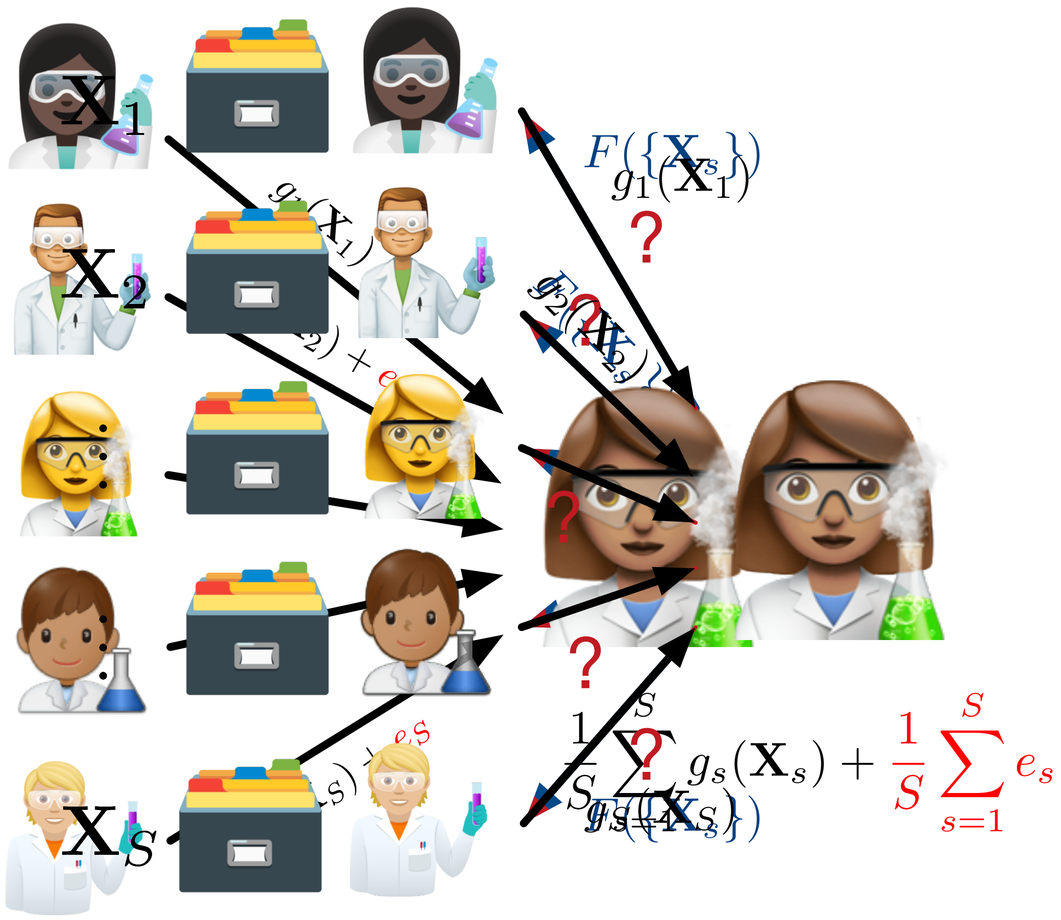
Federated computation in the “multi-silo” setting can support many tasks:

- Basic statistical estimation.
- Model training and machine learning.
- Data visualization for exploratory analysis.

Offers a rich set of tasks in which to see what happens to DP between pseudocode and system considerations.

Multi-silo federated learning with privacy

Distributed gradient descent and differential privacy



Regular federated learning:

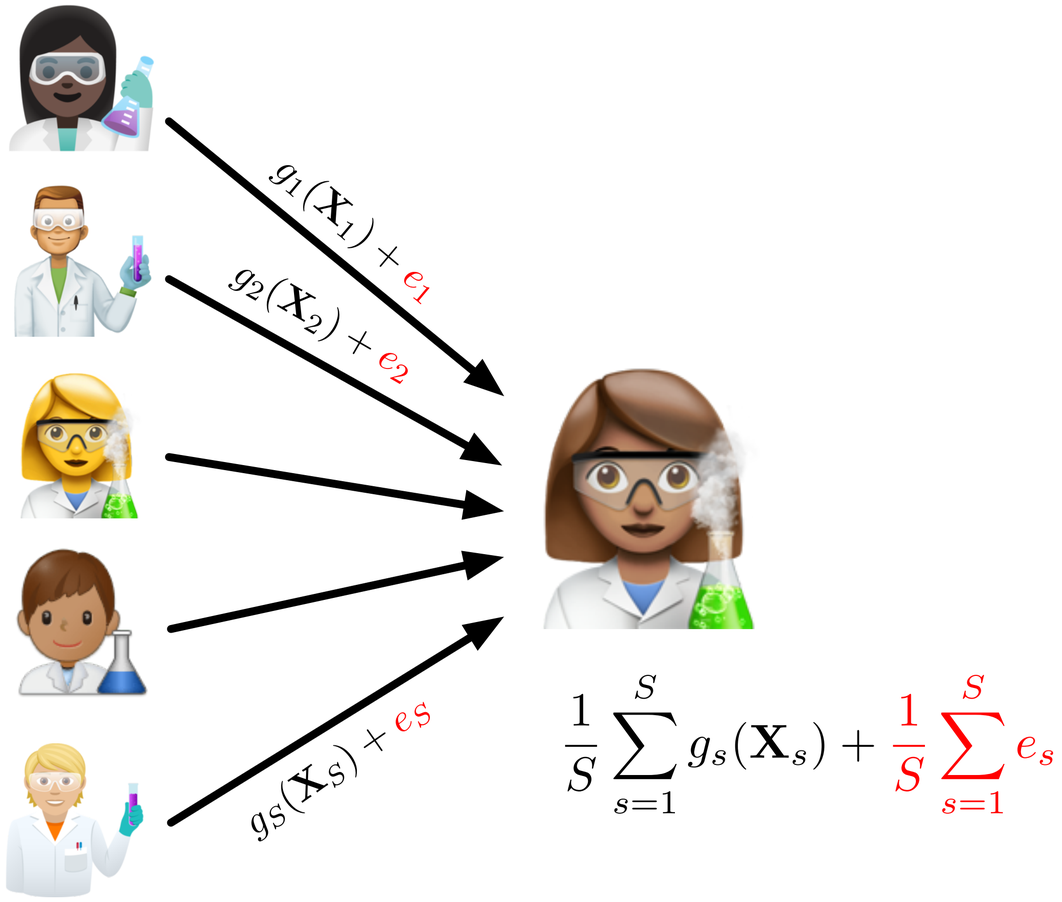
- Form a consortium.
- Sites send gradient updates for a common model.
- Aggregator shares back a consensus gradient update.

With DP:

- Make gradient updates noisy.

But wait… it’s not so simple

Privacy can make existing challenges harder

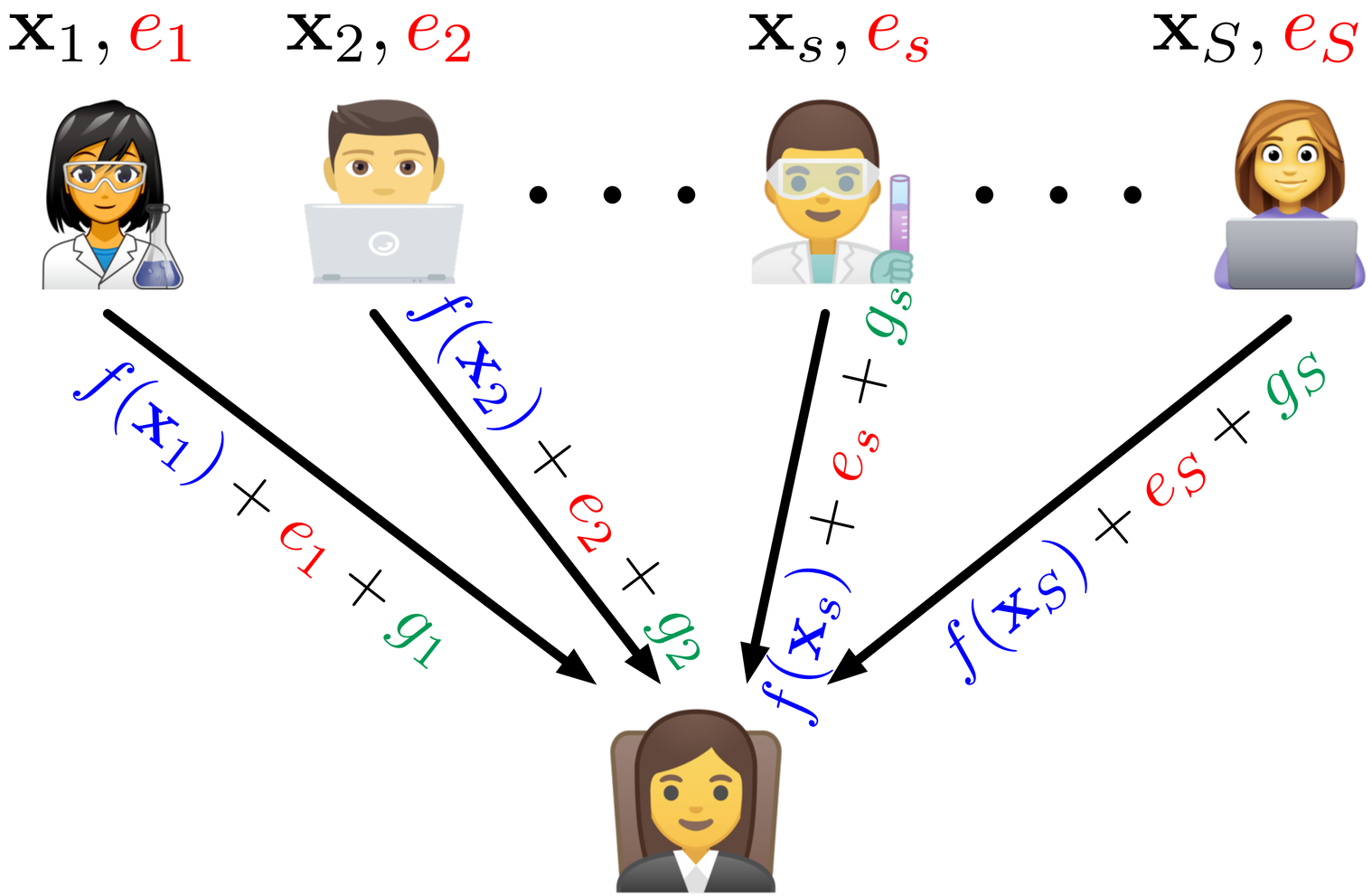


$$\frac{1}{S}\sum_{s=1}^{S} g_s(\mathbf{X}_s) + \frac{1}{S}\sum_{s=1}^{S} e_s$$

- **Gradients may have high sensitivity:**
  - ➡️ clip the gradients
  - ➡️ introduces extra bias
- **Usually requires several epochs:**
  - ➡️ use advanced composition rules
  - ➡️ implementation has to be “exact”
- **Sites can be very different:**
  - ➡️ need to mitigate source variability
  - ➡️ calibration also costs privacy risk

Getting by with a little help from a friend…

Trusted noise generation can help improve performance



Suppose we have a trusted "noise broker:"

- Sites can generate independent noise.
- Broker lets them *anti-correlate* part of the noise.
- Part of the total noise in DP federated learning will cancel out.

Can get close to centralized performance!

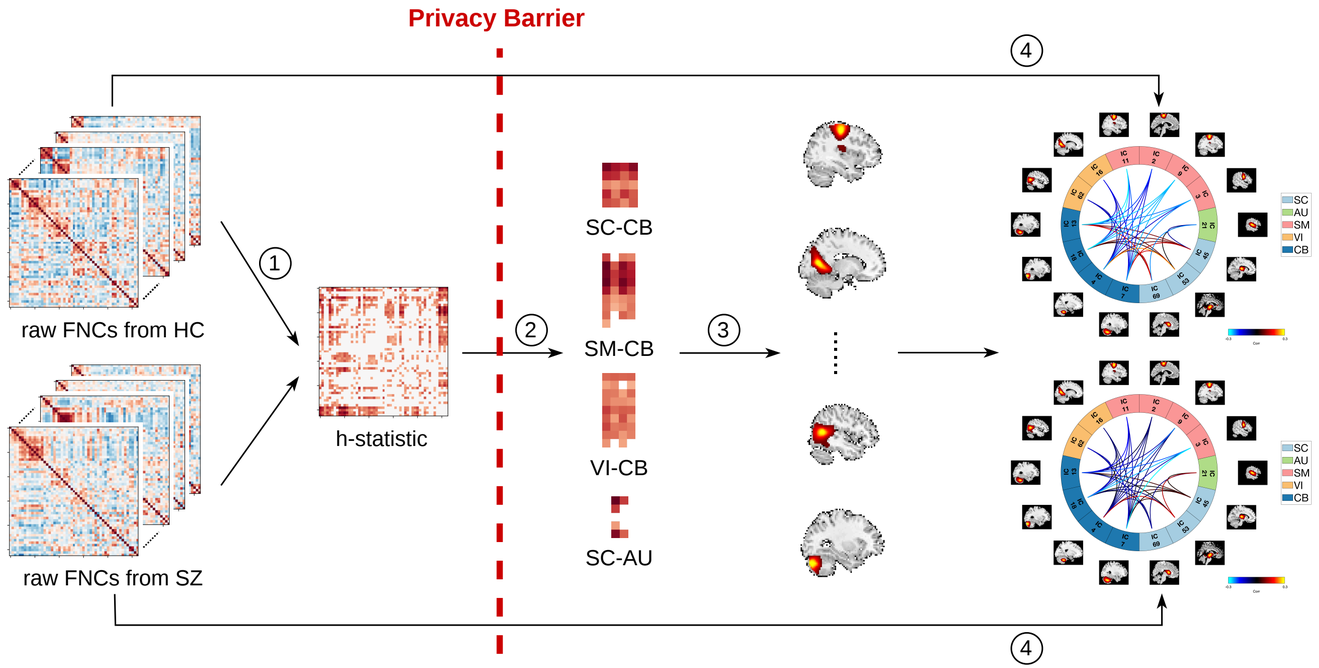
[Imtiaz et al., 2021]

Data visualization with differential privacy

Quantitative artifacts used qualitatively

Visualizations can be *blatantly* non-private: scatter plots. DP visualization raises many new issues:

- Multi-stage workflows: where do we randomize for privacy?
- Are some visualizations easier to privatize than others?
- How does *visual utility* differ from quantitative utility?

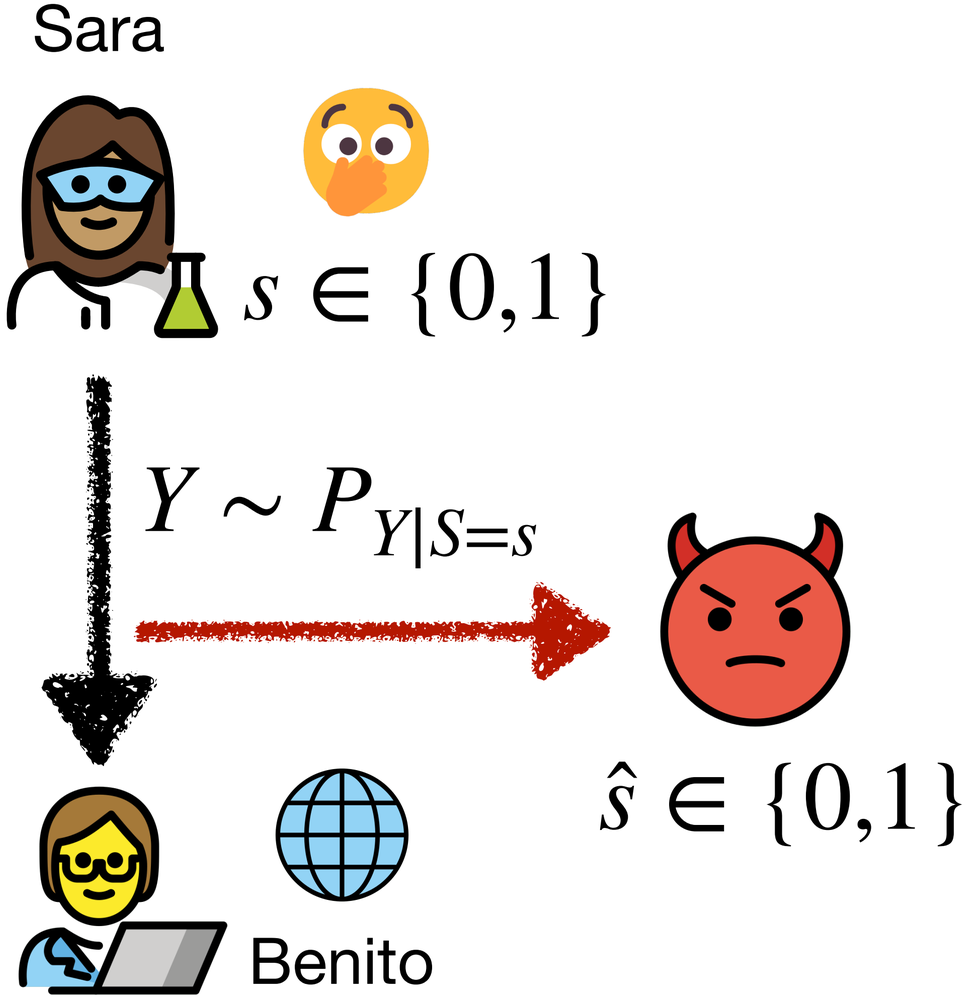


[Tao et al. 2026]

moving forward

Where do we go from where we started?

Wrapping up the end of the tour



We started out with a simple story: **protecting a single bit**.

- Differential privacy both is and is not just as simple as **hypothesis testing**.
- Taking an **statistical signal processing view** opens the door to better analyses/composition.
- The **gap between algorithms and analysis** is shrinking.
- The **gap between algorithms and applications** is still large.

The gap between theory and practice

It's wider than you might think

There are lots of issues with implementing differential privacy in practice. **Most DP implementations are cutting corners.**



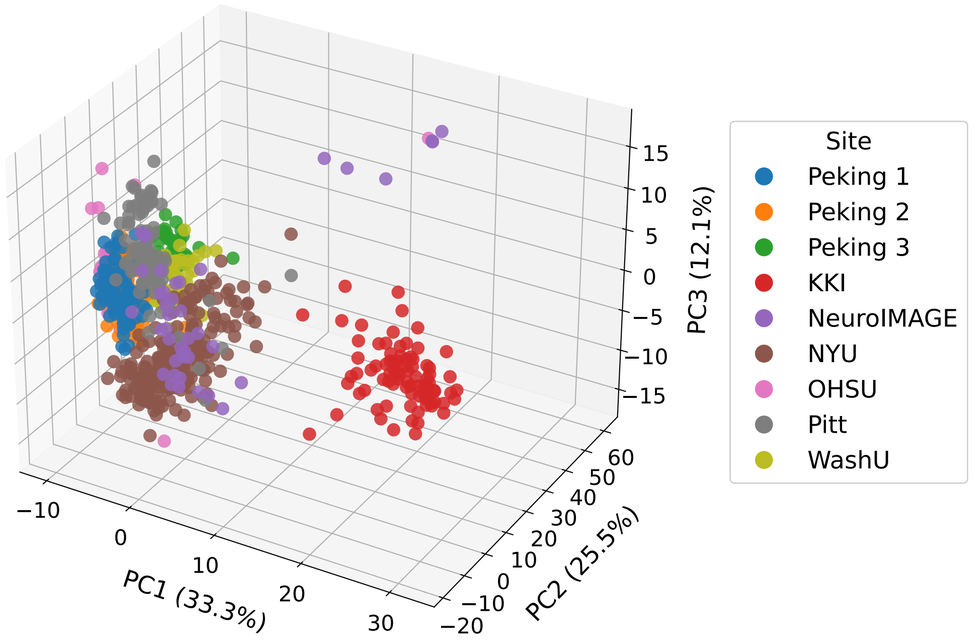
- Approximate versus exact sampling (and side channels)
- Approximate versus exact optimization
- "Privacy amplification" and it's implementation
- Numerical precision and floating point problems
- Managing privacy budgets

The OpenDP project is trying to address some of these issues.

OpenDP

More translational research needed

Implementations reveal new fundamental research questions



[Opinion] The case for “practical” DP has not been made (yet).

We have had 20 years of differential privacy. What will drive future development is *translational research*.

- An opportunity to connect DP to existing (known) challenges.
- Many DP theory/algorithm advances have been driven by applications!
- DP is not always the right answer: SP applications can drive new privacy frameworks.

Thanks!